

Universe24: an integer cellular automaton read through clicks, and the formulas it shares with nature

Alon Gonen

Independent researcher, Haifa, Israel

alon@defounder.ai ORCID 0009-0008-2701-4599

October 2026

Abstract

How much of physics can one local rule in whole numbers compute, with nothing continuous beneath it and a click, the model's name for what a detector registers, as the only measurement? We present a method, Universe24: an integer cellular automaton, a cubic lattice of Nodes stepped by one recurrence, which takes a Node's present, its past and its six neighbours through one integer division. The step is reversible, conserves a quadratic form exactly in the rationals, and is covariant under the 48 symmetries of the cube; no boost is among them. A change crosses at most one Link per interval. Light moves at $1/\sqrt{3}$ of that bound, and nature's c sets the length of a Link and of an interval. The lattice is read only by NodeReaders, declared regions of two Nodes or more that read their Nodes and write the clicks that happened; a NodeReader is the gate between the lattice, which computes, and Universe24, the world of the model, which is those clicks. Measurement is a meeting at a NodeReader: the record run forward from its root meets the NodeReader's own transition read backward from the click that happened, two quanta and never one. The NodeReader draws the click; the lattice draws nothing. That draw is the one non-local element. On this basis, the wave equation, the quantum of action and the metric of gravity are three readings of one line. The line gives its own formulas with no number taken from nature. A family's pair $[p, q]$, with $\cos \omega_0 = p/q$, gives the family's inertial mass $m^* = 3 \tan \omega_0$ and its light speed $c_m^2 = \omega_0/(3 \tan \omega_0)$; $E = m^* c_m^2$ holds exactly, and Lorentz's factor holds at c_m as the gap closes. The matter pair gives the post-Newtonian $\gamma_K = q/p$ and $\beta_K = (q + p)/(2p)$, which go to 1 as the gap closes. It also gives charge universality, the ratio $(m_p/m_e)^2$ between the two forces across families, and $\alpha = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$ between two quanta, k a weight declared in the file. Newton's gravitation comes out, G a reading, and Einstein's weak field follows under one named assumption fitted to two clicks: the metric is exponential, $\gamma = 1$ exactly, $\beta = 1$ as the gap closes, and the shadow is 4.6 percent larger than Schwarzschild's. The paper claims the method and not nature: Universe24 gives formulas that also appear in nature, each comparison a supporting finding for the reader to test, and what the method does not give is listed.

Keywords: integer cellular automaton; reversible lattice dynamics; measurement as a meeting of past and future; lattice dispersion and inertial mass; weak-field gravity on a lattice; limiting speed on a lattice

1 Introduction

The question. This paper is about one method of computing physics. A GameBoard of Nodes is stepped by one local, reversible rule in whole numbers, and it is read only through clicks, the one kind of measurement there is. In the words of the title: the integer cellular automaton is the GameBoard stepped by Rule3; Universe24 is its clicks, the records and clicks of its declared files read through NodeReaders; and the formulas it shares with nature are the comparisons of Sections 8 to 10. Those comparisons are supporting findings, never the claim. On this GameBoard the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, Rule3. The line itself is the lattice wave equation. Its count, read in quanta, is the click. Its paces, read as a clock and a ruler, are the metric. Nothing continuous lies beneath. The question is whether this method can give the formulas that nature's measurements show, and how little state, non-locality and randomness it needs. The fifteen formulas, the one computed file set beside its blind row, and the two gates below are examples of what the method yields. They are not its scope, and Section 10.3 lists what it does not yet give. The paper's one claim is the method: a GameBoard of integers, stepped by one rule and read into the world through clicks, from which the forms set beside nature's are derived in the order stated. The paper does not claim that nature is such a lattice. Every derivation, comparison and prediction below is a supporting finding for the method, each with its status, and none is the paper's thesis. The paper computes the click patterns of declared NodeReaders and sets them beside nature's measured patterns. The paper starts from one statement and never contradicts it (ASSUMPTION): Universe24 is the meeting of the past with the future; a click is that meeting, the one measurement there is; and in it only clicks are seen. The clicks are the mechanism's own. In an experiment of nature a detector, of whatever kind, registers something, and the model's click stands for that registration; but the click is defined by the model, one quantum credited to a NodeReader, and not by nature, so the two are not the same thing, and the paper compares what its clicks give with what nature's detectors register. Every physical number in the paper is a reading of one integer rule with its declared integers: the pairs, Γ , T , the two holders' weights, gravity's E_g , the sign holder's k , and the vacuum content (Section 6.6). For nature's other clicks it gives the forms of Section 8 at the computed pair, with the findings of Section 10.3 set against them. A Node holds a record's present and its past, two integer levels. The rule makes the record's future from these two and from the six neighbouring Nodes. The rule is covariant under the translations of the lattice and under the symmetry group of the cube, which has order 48; no boost is among these symmetries (Section 3.1). The lattice holds nothing but events that split: at every Node, a level splits by the rule to the six neighbours. Run forward from the root, the lay, the split events are the future, every path at once. Read backward from a click that happened, by the same rule, they are the past, and through two slits both of them pass through

both gaps. A click is where the two meet at a declared NodeReader. Which meeting happens is the one thing the model does not determine, and it is the one non-local element of the description (Section 5.2). What looks like a wave is the future. What looks like a particle is the present, the click. The track is the past (Section 5.1). No theorem of minimality is claimed.

Why the world is read in clicks, and never on the GameBoard. That measurement is a click follows from what the model is. Whoever sees is itself a cluster of bodies on the GameBoard, and a Node reads its six Ports and nothing else. Inside a cluster everything is seen: a GameBoard reading is the local language of the cluster's own record. Beyond the cluster's tail nothing reaches it except what a NodeReader reports, one quantum credited at a meeting; a click is the only thing that passes between a cluster and its neighbour [1, the observer is a cluster; the click is the pointer]. What stands on the GameBoard between clicks is not a fact of the world but its possibilities: the split events of every path at once, the levels Rule3 makes at the Nodes, the future. A fact is a realised event, a quantum credited to a NodeReader, and that happens only at a click. So in Universe24 there are clicks and nothing else, and nothing of the GameBoard is measured, only its clicks through a NodeReader [1, no waves on the event board; what produces a click is not known]. This is why the paper sets beside nature only what the clicks give: a GameBoard reading, a level, a support, a count at a Node, is a diagnostic of the computation and is labelled so, and only a click pattern is compared with a measured pattern of nature [1, readings and measurements] (ASSUMPTION, the first postulate of Section 3.6). So the paper has three words and keeps them apart. The GameBoard is the lattice stepped by Rule3; it computes. The NodeReader is the gate between the GameBoard and Universe24: it reads the board and writes the clicks that happened. Universe24 is those clicks, the world of the model. The paper's one claim is the method. Universe24 gives formulas that also appear in nature, set beside nature's by supporting findings and never by proof, and how close they come is for the reader to judge from them.

The rule in one line. Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write, three readings, the declarations of Section 3 and the one assumption of Section 3.4. The line holds at every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

Rule3: $w a_{\text{next}} + r' = \sum_{j \sim i} R_{ij} \text{arr}_j + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_{ij} = 2 \text{ num } p_a(i, j)^2 = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - \sum_{j \sim i} R_{ij};$ the one write: a held part at a Node gains $\left(\sum_f w_f q_f(\text{Node}) + r \right) \text{div } E$, the remainder r kept there; the readings: $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i, \quad W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$	(1)
--	-----

In the first line, a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval later: the present, the past and the future. arr_j is the arrival through the Port to the neighbour j , which is that neighbour's level; the six Ports are summed. r and r' are the remainder kept at the Node before and after the step, and Γ is the Node clock. $[\text{num}, \text{den}]$ is the family's pair, and $\cos \omega_0 = \text{num}/\text{den}$ is its rotation per interval at wave number zero. p_0 is the Node's clock pace, p_i its own pace, and $p_a(i, j)$ is the pace of the Link through the Port: the Node's pace times the Link's factor q_{ij}/Γ . The factor reads the same from both ends; the pace is the reading Node's own. All of these are the Node clock, lowered by the holders' levels that the family reads there (Section 3.4). At the vacuum's paces, the two Links of an axis share one $R_a = 2 \text{ num } \Gamma^2$, and $S = 0$. div is floor division. The second line is the only way a holder's level is ever written. It sums, over the sourcing families f , each family's own quantity q_f at the Node, weighted by w_f ; adds the part's one remainder; and divides by the part's wall E . The third line is what is read from a record and never written: its form D_i , which sources the holders; its Wronskian W_i , which is its charge; and its current F_{ij} into the Node i from its neighbour j . A NodeReader reads that current through its boundary in quanta of $W_c = 3 \text{ den } T$, where T is the quantum's norm, the normalisation of the quadratic form.

The three layers of the method. The method has three layers, and each has a fixed part and a free part (Fig. 1). The first layer is the board's rules (Section 3): which families exist and of what dimension, which holders, who reads what and how. Here Rule3 with its reads is fixed, and the rows of the file are free. The second layer is the numbers (Section 6): Γ , the pairs, the holders' weights, gravity's E_g and the sign holder's k , T , the vacuum content and the ranges. All of these are free, and there are as few of them as the rule allows. The third layer is the click (Sections 5, 5.3 and 7.4): how the board becomes a click, what the NodeReader declares, what is credited, and how a mass on the board is read as a mass in clicks. Fixed here, and common to every file, is the one form of the click: the meeting, the bilinear credit, the draw by the shares, the NodeReader's one write at the Node of the exchange, and the postulate that only a click is a measurement. Free, and belonging to the file, are the NodeReader's four declarations, among them the unit it counts in.

The formulas derived. The following formulas come from Eq. (1), in the order of the paper. Each has its mark in Table 1 and its algebra in the supplement. Light's band, its dispersion and its index in a well (Sections 3.3 and 3.4). The lattice Klein-Gordon equation and its Schrödinger limit (Section 3.3). The conserved form and the share identity (Sections 4.3 and 4.4). The Wronskian as the charge, and the rests of the holders (Sections 5.5 and 6.4). The functional of a bound body and its stability (Section 7.2). The adiabatic invariant and the drift of the count (Section 7.4). The rate of a clock, the bending of light, Shapiro's delay, the perihelion and the capture radius of the shadow, the moving clock, de Broglie's and Planck's relations and $E = mc^2$ in the band's form, the index of the content for every family, and the fall as refraction (Section 8). Bell's E and S , Tsirelson's bound and GHZ's M under the declared credit (Section 9.4). The fine-structure constant as a coefficient of the file, the hierarchy of the two forces between families, the units and the highest photon energy, and Newton's G as a reading (Section 10.1). The formulas of Tables 1 and 2, several with their companions, cover five areas, and every one is an algebraic consequence of the line and of the inputs named beside it.

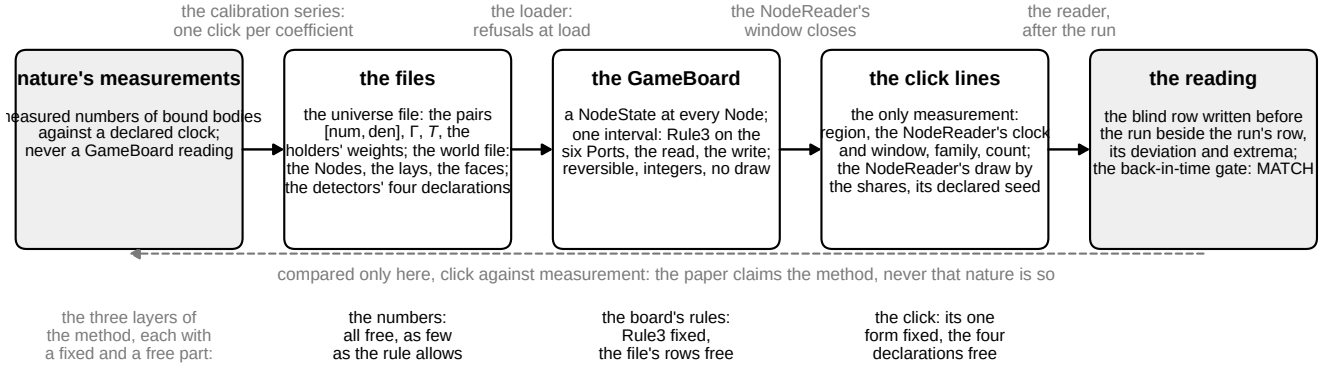


Fig. 1 The method, from nature's measurements to the paper's reading (Sections 1, 3.7, 5.3 and 10.2; drawn from the definitions, no run). Nature's measured values on bound bodies are read into the integers of the files, one click per coefficient; this is the calibration series. The files declare Universe24. The GameBoard is stepped by Rule3, one interval of acts on the six Ports, reversible, in whole numbers and with no draw. The NodeReaders write the click lines, the only measurement. The reader prints the row of the run beside the blind row, which was written before the run, with the MATCH of the back-in-time gate. Only there, click against measurement, does the computed Universe24 meet the formulas of nature it is set beside. The paper claims the method, never that nature is so.

What the paper contributes. We state the rule and what is exact about it, with proofs in the supplement: reversibility to the bit, the conserved form with its current, and the momentum flux as the wave's own stress (Section 4). We formulate measurement as the meeting of the forward record and the backward record at a declared NodeReader, and we name the one non-local element (Section 5). We obtain the families as bands of the rule, and we choose a universe with as few free numbers as the rule allows (Section 6). We show that bodies are fixed points, prove their stability under dilation, and state how they cluster (Section 7). We compute the weak-field forms of gravity and the factor of the moving clock, each marked by what it rests on, both in the model's own potential and in the potential that nature's tests read (Section 8). And we state the two slits and Bell as derivations of what each must give; neither experiment is performed here (Section 9).

What stands, and on what. Five things, each with its mark. (i) The core. Rule3 and the boxed lines after it, the paces, the conserved form, the share, the Wronskian, Kepler's potential and G as a reading are theorems of the line, or are derived from it with no number of nature in them, and the step is reversible to the bit (Section 4). What the checks at this paper's closing corrected lies beside them, in secondary derivations, marks and digits, and not in the line. (ii) One rule in integers reaches Einstein's weak field. From the composed paces alone, under the one named assumption of Section 3.4, come the exponential metric, the bending $4U_b$, Shapiro's delay, the redshift and the push $E(1 + \beta^2)\nabla U$, with $\gamma = 1$ exactly in the clock's potential and $\beta = 1$ as the gap closes. The assumption is fitted by two of those clicks, and the others are its predictions (Section 8). (iii) The click is the one measurement, and the non-locality is accounted for in one place. Bell's 478/169 and the GHZ's -4 follow from the declared credit; the two slits' 278 stands against the blind 273; the one non-local element is the credit's one draw through the root. (iv) Every claim carries its mark and its fence, and what fails is named with its number: $\alpha_1 = -8$, hydrogen's $1s$ at -0.043 of Rydberg's unit under the self-read (S.46), and the telegraph of the shelved ion (Section 10.3). Nothing is claimed beyond its mark. (v) Everything can be recomputed. The files, the engine, the scripts and the documents are public (Section 3.7).

What is put in, and the three kinds of claim. The paper's claims are of three kinds, and the text keeps them apart. *Found*: what is proved or derived from the rule with no input chosen for it. These are the fifteen formulas above and the click formulas of Table 2 marked derived. The findings against nature are comparisons, listed in Section 10.3. *Assumed, chosen or declared*: the assumption and the credit below, the rows of the universe, the NodeReader's four declarations, among them its unit of credit, the electron as the free quantum of matter, and the denominator of the matter pair as the family's own integer (Section 10.1). *Held to be computable and not yet computed*: the bodies at the closing gap, the moving fixed point of a body, and the absorbed fraction with the body's write. The explicitly named hypotheses are stated where they stand; among them are light's gravity, entered as a derived hypothesis (Section 10.2), the carrier that would mend the lattice's frame, and the nuclear holders (Section 10.3). One assumption is named with the forms: a Link is clocked twice by the clock of the Node that reads it, $hN = 1$, fixed by the bending and the perihelion (Section 3.4). The forms against Kepler's potential reach Einstein's only as the gap of the matter family closes (Section 8). One choice was made with a known result in view, and we state it as an input, not a prediction: a click's share is the square of the meeting. This credit was chosen to reach nature's correlations, so Bell's $2\sqrt{2}$ follows from it and does not test it. The numbers of the file are nature's and not the rule's: the vacuum content, a declaration; the rate of the vacuum write, Λ , the one declared number of the hypothesis of Section 7.5; the binding range, a file's choice; gravity's weight E_g and the write weight k of the sign holder, read from G and from α ; and the gaps, read from the masses (Section 10.2, Table 4). Table 1 lists every result of the paper with its kind, its fence and its inputs. What is not claimed: that nature is a lattice; that any number of the computed pair $[2, 3]$ is nature's; that a comparison with nature is more than a consistency statement at the stated order; or that a formula known in the continuum is new where the paper gives its lattice form. Every derivation was checked twice and independently within the project, and, where a file exists, by the checks of the implementation. No independent referee has checked them. Where the text says exact, it means exact in the rational arithmetic before the rounding, with the integer remainders bounded as stated. A line decided after a derivation is stated as the law, with the run of the implementation

pending; a run that differed from its prediction would be a defect of the implementation to repair, never a result.

How a Universe24 file is built and tested. A Universe24 file declares the following, and nothing else: a family's pair and dimension; a holder's sources, divisor and rest; the bodies and the messages; a NodeReader's four declarations, which are its region, window, floor and whole, and its seed; and the faces. The engine holds no number, no formula, no family name and no flag, and every primitive is one folder found by its name [2, how to run a world; how to add a family]. One change, in numbers the paper keeps, shows what this means: replacing the matter pair [2, 3] by [999, 1000] moves c_m/c from 0.867 to 1, and moves the computed pair's γ_K and β_K from 1.5 and 1.25 to 1 (Section 8); a binding holder's range R turns a declared body from unbound to bound (Section 6.4). The procedure, with the two slits as the example, is this. The blind row is written from the law's lines before the run. The run is made. The back-in-time gate reports MATCH or MISS on its reversibility. The reader prints the row of the run beside the blind row, with its deviation and its extrema (Sections 3.7 and 9). Three files are run and gated in this paper: the two slits, Bell's gate and the GHZ gate. Every other experiment the paper sets against nature is derived from the law's lines and not run, each with its mark and fence where it stands. These are Einstein's weak-field forms, the bending, Shapiro's delay, the redshift as the clock at two heights and the perihelion at the small gap (S.15, S.19, S.20); the shadow (S.30); the push on a moving record (S.58); the two slits' own blind row, written before the run (Section 9.2); the moving clock (S.21, S.22); the magnetism of moving charges (S.33); the Eötvös ratio (S.47); the one-quantum image of the lens (Eq. (17)); the deceleration of the clusters (S.29); the atom (S.46); Rabi's absorption line (S.43); the gravity of light (Section 10.2); the anticoincidence (S.57); the Zeno effect (S.59); the preferred frame's α_1 and α_2 against Gravity Probe B and the Sun's spin (S.35), derived and failing; and, derived under a declaration, the telegraph of the shelved ion and the giving of an excited body (the rate declared; Section 10.3) and the nuclear binding (S.45, the weight declared). Their engine files are read beside their blinds in Section 10.3, as named findings and not as a gate. Rule3 is fixed and no rule is added. A new line of physics enters as a hypothesis under its own name, after the three tests, generic, vector and local, which is how every family and holder of this paper entered (Section 3.6).

How to read the paper. Every claim carries one of seven marks. THEOREM: proved for every lattice. DERIVED: obtained from theorems by argument. COMPUTED: a number obtained from the rule's lines before any run. ASSUMPTION: a line held by decision and named where it is used. HYPOTHESIS: a line outside the model, under its own name. EXPERIMENT: a number the model does not fix, with the experiment named that would fix it; the paper performs none. FITTED: an input chosen with a known result in view. The status columns of the tables use the same marks, and beside them calibration (a number taken from nature), declaration (a file's own number) and implemented (checked by the implementation). Every formula carries its fence beside its mark. GAMEBOARD: a reading of the lattice's own numbers that no NodeReader reports, a diagnostic. CLICKS: a formula for what a NodeReader reports, the share Q_R/W_c on the forward orbit (Section 5.2) or a ratio of two such shares; this is the only kind that can be set beside nature. The algebraic steps of every derivation that the text states without its proof are in the supplementary material (Online Resource 1), Derivations S.1 to S.62. They are numbered in the order they were added, each names its section, and they are cited as (S. n) where used. Every sentence of the paper that says what a thing is, light, mass, an atom, gravity, the direction of time, is a sentence about the model and its files, and never about nature. Nothing in the paper is a proof about nature: the theorems are theorems about the rule, and every comparison with nature is a supporting finding and not a proof. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3].

The words. The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the lattice of Nodes is the *GameBoard* (Fig. 2(a)). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value, with its pair and with what it holds. Its *record* is its levels over the GameBoard, two per Node with a remainder. A family of quanta has a *dimension*: 1 for one real line, 2 for a plane of two lines. A *body* is the set of Nodes on which a family's record stands bound; it has no law of its own. A *cluster* is bodies standing in one another's wells. A *message* is a travelling record laid by the file. *Universe24* is the world the method computes: the clicks the NodeReaders write from the GameBoard, never the GameBoard itself and never nature; the paper's one claim is the method, and the formulas Universe24 gives are set beside nature's. A *file* declares one experiment of it and the implementation runs it: the families with their numbers, the holders, the bodies and the messages laid, the NodeReaders with their declarations, and the faces; the engine's documents call a file's run a world, a word this paper keeps for the world we live in. A *NodeReader* is a connected region of two Nodes or more declared in the file with its seed; it reads its Nodes and writes the clicks that happened, naming itself and never a Node, and, given a record of its own with parts and a lifetime, it gives as well. Its *click* is one quantum credited to it, the one measurement, the model's name for what a detector of an experiment registers and defined by the model alone. An *interval* is one step of the whole GameBoard. A family that holds a value for others to read is a *holder*. The words in full are in S.49 [1, the objects][2, the words].

2 Related work

Reversible cellular automata and digital physics. The idea that the physical world is a discrete computation goes back to Zuse [4]. Fredkin, Toffoli and Margolus developed it as reversible cellular automata [5][6][7]. Wolfram developed it as a programme of cellular automata [8], and 't Hooft as the cellular automaton interpretation, a deterministic foundation beneath quantum theory [9][10]. Quantum cellular automata put a unitary step on a lattice. They range from the lattice Weyl, Dirac and Maxwell equations [11] and quantum lattice gases [12] to the structure theorems of the reviews [13][14] and the derivation of Weyl's and Dirac's equations from an automaton's symmetries [15]. The step of the present rule is an integer recurrence on two levels, not a unitary on amplitudes. Rule3 is a second-order cellular automaton in the sense of Toffoli and Margolus, $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$ [7]. Its alphabet is the integers, and it has one Euclidean division whose remainder stays at the Node. This is what the title calls an integer cellular automaton. Feynman noted that a local classical computer reproduces no correlations beyond Bell's bound [16]. The model agrees, and it places the excess in the declared credit of the click, not in the lattice (Section 9.4).

Table 1 The results of the paper, each with its kind, its fence and what it rests on. A calibration is a number taken from nature; EXPERIMENT a number the model does not fix.

Result	Kind	Fence	Rests on	Section
Rule3's step is reversible to the bit; the interval under the condition of Section 4.1	theorem	GameBoard	Rule3	4.1
The back-in-time gate says MATCH on the two slits', Bell's and GHZ's files; Bell's and the GHZ's readers return their blinds exactly, $S = 478/169$ and $M = -4$; the two slits' reader $N = 278$ against the blind 273, five quanta over, 0.3 of $\sqrt{273}$ (Section 9.2)	computed, a run	clicks	the implementation	4.1
The rule's acts form no product of levels	theorem	GameBoard	Rule3	4.2
A conserved form, its current, and the momentum flux as the wave's stress	theorem	GameBoard	Rule3	4.3
The count is the record's share; the total share is never negative	theorem	GameBoard	Rule3	4.4
Three exact massive rotations	theorem	GameBoard	Rule3	6.1
Light's speed $1/\sqrt{3}$ Links per interval, its dispersion, and no dispersion along the cube's diagonal	derived	GameBoard	Rule3 for the pair [1, 1]	3.3
A click is the meeting of two quanta, decided in the credit, written at one Node; no signal	assumption, derived	clicks	the meeting, the NodeReader's draw and write	5
A holder's reach from its gap, Yukawa's form	derived	GameBoard	Rule3 for a gapped pair	6.4
The stability of a body under the dilation, to first order in the well	theorem	GameBoard	the body's functional	7.2
A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$, the count's drift	derived	clicks	Rule3 at the body's Nodes	7.4
A cluster: the three levels at a Node as the nesting, three kinds of binding, the click the only thing between clusters, the hierarchy recursive	assumption, derived	GameBoard, clicks	the universe is bound; Rule3	7.5
The gravitational redshift, $z = f(\omega_0) U$	derived	clicks	the clock's read	8
The bending and the perihelion, which fixed the one assumption; the delay, the redshift's factor and the shadow's capture radius $2e m$, its predictions	derived under the one assumption, fitted by the first two	clicks	a Link clocked twice by its Node's clock, $hN = 1$	8
The same forms against Kepler's potential: γ_K , β_K and α_K from the pair alone, closing with the gap	derived	clicks	the fall's formula	8
The moving clock: its fourth order and its kinetic scale c_m , Lorentz's as the gap closes	derived	clicks	Rule3's band	8
The force between moving charges, Coulomb's over γ	derived; the implementation's run a check of its bookkeeping	clicks	the rule's continuity identity, the sign's covector	8
De Broglie, the fall, Newton's pull; the content's index for every family, the lens	derived	GameBoard, clicks	the band and the holder's rest	8
Newton's pull, the clock's G and Kepler's G	derived	GameBoard, clicks	Rule3 for [1, 1] and E_g	10.1
$\alpha_{\text{law}} = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ between two quanta, k the sign holder's declared weight; charge universality; the two forces' ratio between families $(m_p/m_e)^2$; the units; the share's ceiling; the gap below 10^{-19}	derived; k declared	clicks	the rotation, Poisson's rest, de Broglie; the quantum of a family; LHAASO	10.1
The two-slit fringe row and the lattice's arrival interval	computed	clicks	Rule3 on the declared file	9.2
Bell's $E = \cos 2(a - b)$, $S = 2\sqrt{2}$, $S(\rho)$ and Tsirelson's $2\sqrt{1 + \rho^2}$; every local credit at most 2 (theorem)	fitted; derived	clicks	the credit rule, the meeting's square	9.4
The vacuum content 0; gravity's events at light's speed in every level	assumption; implemented	clicks	declared; every row reads the content	6.5
The binding range, 20 Links	a file's choice	GameBoard	declared, a reading of an earlier version the families replace	6.4
Gravity's weight E_g and the sign holder's k , the matter pair's gap, the Link's length	calibration, experiment	clicks	nature's measurements	6.6, 10.2
The two clocks shift alike; the findings against nature of Section 10.3	derived; comparisons	clicks	the principle; Rule3	10.3

Time-symmetric accounts of measurement. Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]. Cramer's transactional interpretation has an offer wave forward and a confirmation wave backward, and they meet at the absorber [18]. The two-state vector formalism evolves one state forward from the preparation and one backward from the outcome [19][20]. Price argued that a time-symmetric account is what Bell's theorem leaves open [21]. Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class: the backward reading from a realised click depends on the NodeReader's setting alone. It differs in three respects. Both directions are one integer line, exact to the bit; their product is a reading and not a wave; and the choice among the meetings is the one non-local element. The record is an ontic state in the terms of [23].

Bell's theorem and local models. Bell's inequality bounds every local account of a pair's correlations [24], in the form that experiments test [25]. A local model, a joint distribution and the inequalities are one condition [26]. The model's local credits give $S = \sqrt{2}$ and $S = 2$, as they must (Section 9.4). Its credit by the contrast reaches the quantum correlation only by discarding pairs. This is Pearle's road [27], and it runs below the efficiency bounds of Eberhard and of Garg and Mermin [28][29]. The loophole-free tests of 2015 close that road for nature [30][31][33].

Discrete gravity and its tests. Regge calculus [35] and causal sets [36] discretise the geometry of space-time itself. Here the lattice is never curved, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework [38], the one assumption gives $\gamma = 1$ exactly and $\beta = 1$ at the small gap, in the model's own potential. Against the potential that the tests read, the matter band's factors close as its gap closes. A lattice breaks Lorentz invariance at order one unless its terms are tuned [39]; the bounds are reviewed in [40]. The paper's answer is no loops and the Link's bound of S.24, with the closing gap for matter (Section 8).

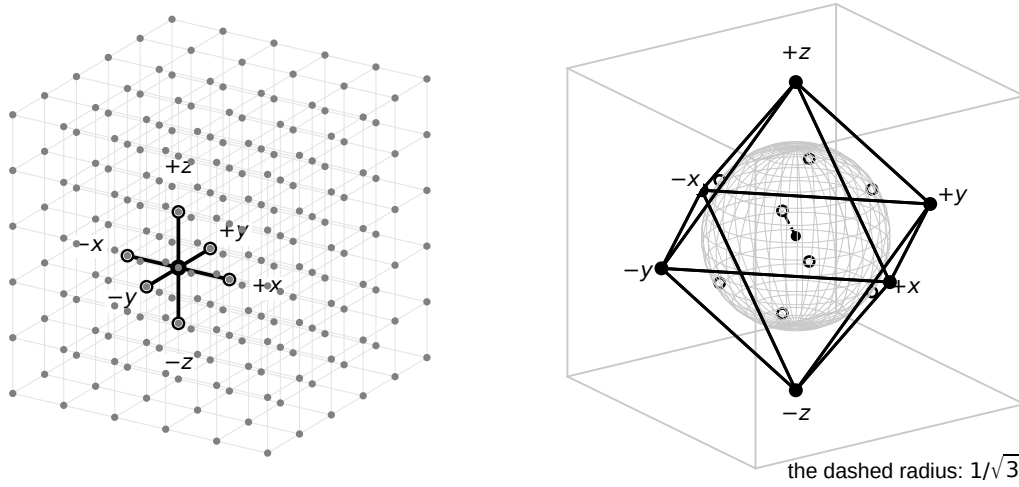


Fig. 2 (a) The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines). One Node and its six neighbours are marked; these are the Nodes one interval away from it. A Node keeps each family's two levels and its remainders, and nothing else. (b) The Nodes one interval away from a Node form the octahedron $|x| + |y| + |z| \leq 1$, whose six vertices are the six Ports. Its symmetry group is the cube's, O_h : the 24 rotations and, with the reflections, the 48 signed permutations of the axes. Rule3 is covariant under this group (Section 3.1). The inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the diagonals of the cube, and that radius is light's speed in Links per interval (Section 3.3). Both are drawn from the definitions; no run.

3 The model

This section states the model: the lattice, the families, the rule, the paces by which one family acts on another, the one write, the readings and the order of acts in an interval [1, the GameBoard and the families; Rule3; the primitives].

3.1 The GameBoard

The GameBoard is a box of $N_x \times N_y \times N_z$ Nodes. Two Nodes one step apart along one axis are joined by a Link. Each Node has six Links, one through each of its six Ports, $\pm x, \pm y$ and $\pm z$. Each axis is declared periodic, open or folded. Beyond an open face there is no Node, and a level there is read as 0. A folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps; this is how a wall with slits is made. A face may also be declared receding (ASSUMPTION). The box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face. So no event ever meets the face, and a run on such a box is a run on the unbounded lattice. The box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board][3, the receding face].

The group and its action. The symmetry group of the cube has 48 elements: the 24 rotations and, with the reflections, the signed permutations of the three axes. It is O_h , the group of the octahedron, which is also the group of the cube (Fig. 2(b)). With the translations it is the space group of the simple cubic lattice. It acts by permuting the Nodes and their Ports, and it carries every level pair with its Node. The time reversal exchanges a record's present and its past and carries the step to its inverse (Theorem 1). Rule3 commutes with the action of the space group, since its line reads the six Ports symmetrically and names no axis. Under the group, the form D is a scalar, even under the time reversal. The current F is a vector, odd under it. The Wronskian W is a scalar under the cube's 48, odd under the time reversal and under the plane's reflection. The click is a scalar: it names a region and an interval, never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group. So Lorentz's group and the continuum's rotations are no symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (Section 8) (THEOREM for the covariance and the kinds of the readings; GAMEBOARD; S.4) [1, the direction; the four acts][3, the one algebra].

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder; and for every part of a holder, the remainder of its one write. A count, a current, a charge or a tension is a reading, computed from these numbers when it is needed. No register, table, formula or random draw is kept at a Node [1, the objects][3, nothing at a Node but the law's numbers].

3.2 Families and records

A family is declared by two keys: its pair [num, den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and it holds nothing. A family of dimension 3, three real lines with a rotation among them on each Link, is declared by no file. It enters only as an explicitly named hypothesis of Section 10.3 (HYPOTHESIS). A holder declares its sources, the form, the form with the three tensions, or the Wronskian, together with its divisor E_s and, for the massless holder, its rest. The sources fix its parts, one real line per source. A holder sourced by the form or the tensions holds the content, which slows the clocks of the families that read it. A holder sourced by the Wronskian holds the sign, the charge's holder. A family's record is its levels over the GameBoard: at every Node a level now and a level one interval before, with a remainder. A family of dimension 2 lays two level pairs (re, im), two solutions of the same line, and only a plane carries a charge (Section 5.5). A body is a connected set of Nodes on which a family's record stands bound, and a message is a travelling record laid by the file [1, the objects].

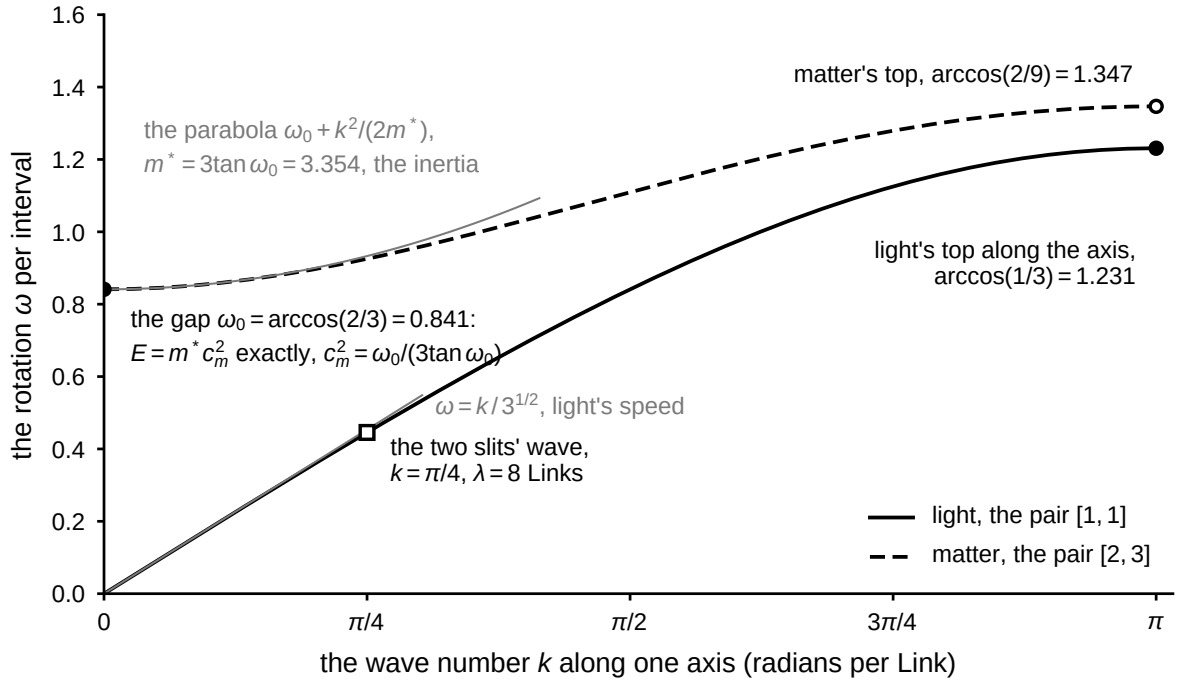


Fig. 3 The bands of light and matter along one axis, from the law's line (DERIVED, GAMEBOARD; drawn from the formula, no run): $\cos \omega = \frac{\text{num}}{3 \text{ den}} (2 + \cos k)$ at the wave number k along one axis, for light's pair [1, 1] (solid) and matter's [2, 3] (dashed). At $k = 0$ the gap is $\cos \omega_0 = \text{num}/\text{den}$, $\omega_0 = 0.841$ for matter, with the parabola $\omega_0 + k^2/(2m^*)$ of the inertia $m^* = 3 \tan \omega_0 = 3.354$ (grey). At $k = \pi$ the top of the band is $\arccos(\text{num}/(3 \text{ den}))$, 1.231 for light and 1.347 for matter. Light's slope at the origin is its speed, $1/\sqrt{3}$ Links per interval (grey). The wave of the two slits, $k = \pi/4$ (the square), sits on light's band at $\omega = 0.4456$, with the period 14.1 intervals (Section 9.1). $E = m^* c_m^2 = \omega_0$ exactly, with $c_m^2 = \omega_0 / (3 \tan \omega_0)$. Along the cube's diagonal light has $\cos \omega = \cos k$, no dispersion; it is not drawn.

3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). One whole number is formed: the six arrivals through the Ports, each weighted by the coefficient R_{ij} of its Link, plus the Node's own level times S , minus w times the level before, plus the remainder the Node kept. That sum is divided once by the wall w . The quotient is the next level, and the remainder, $0 \leq r' < w$, stays at the Node and never travels. Nothing is rounded away, and the step back is the same division. This is why every run is exact and reversible to the bit (Theorem 1). The name comes from the three axes: at the vacuum's paces one coefficient per direction serves its two Ports, and the wall's factor 6 is the three directions twice; in a content the six Ports carry six coefficients, Eq. (1). For light the next level is the sum of the six arrivals over 3, minus the level before [1, the line].

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$. At wave number zero $\cos \omega_0 = \text{num}/\text{den}$. For the massless pair [1, 1] along one axis $\cos \omega = (2 + \cos k)/3$ (Fig. 3), so $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$. Along the cube's diagonal $\cos \omega = \cos k$ exactly, with no dispersion (DERIVED, GAMEBOARD; S.1). Light's speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, below the causal bound of one Link per interval. Its group speed falls at shorter wavelengths as $1 - 0.083 k^2$ [1, the lattice constants]. In nature's names the vacuum's line is the discrete Klein-Gordon equation, exact but for the remainder term. With $S_6 = 6 + \Delta_{\text{lat}}$ it reads $a_{\text{next}} - 2a_{\text{now}} + a_{\text{before}} = c_K^2 \Delta_{\text{lat}} a_{\text{now}} - m^2 a_{\text{now}} + (r - r')/w$, with $c_K^2 = \text{num}/(3 \text{ den})$ and $m^2 = 2(1 - \text{num}/\text{den})$; the rest rotation is the mass. (c_K is the coefficient of the equation. The band's kinetic scale c_m^2 of Section 8, (e), is $\omega_0/(3 \tan \omega_0)$, and its largest group speed is $c \cos \omega_0$. The three are one as the gap closes.) Its slow limit at small wave number is Schrödinger's equation with the inertia $m^* = 3 \tan \omega_0$ and, in a content U , the potential $\omega_0 f(\omega_0) U$ of Section 7.4 (DERIVED, GAMEBOARD). And $m^* c_m^2 = \omega_0$ exactly for every pair. This is the model's $E = mc^2$ at the band's kinetic scale, read in three click numbers; nature's is its small-gap limit. The departure is $\omega_0 / \tan \omega_0$, which is $1 - \omega_0^2/3$ to the order ω_0^2 (DERIVED, CLICKS; S.18, S.21). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, whose dispersion is the known one [41]. What the law adds is one rational [num, den] that fixes mass and speed together, an equation that is exact, departures that are predictions with bounds (Section 10.2), the integers with the remainder, and the paces (DERIVED).

3.4 The paces: how a holder enters a record's paces

A holder enters a family's paces by changing its paces at a Node, which are the coefficients of Rule3 there:

$$\boxed{p_0 = \Gamma(1 - 1/\Gamma)^c, \quad p_i = \frac{p_0(c_i)^2}{\Gamma}, \quad p_a(i, j) = \frac{p_i q_{ij}}{\Gamma}, \quad q_{ij} = \Gamma - t_a(i, j), \quad c = \sum_{\text{reads}} \text{weight} \times \text{the read family's time level},} \quad (3)$$

Here c_i is the content the family reads at the Node: the sum, over its reads, of the weight times the level of the read holder. (The symbol c names the content in this section and light's speed elsewhere.) The clock p_0 composes: each unit of the level slows it by the same factor $1 - 1/\Gamma$, to the unit. The Node's pace p_i is the clock twice, p_0^2/Γ , so that the ruler goes with the clock. The Link's tension $t_a(i, j)$ is one number per Link, from the parts at its two ends (Section 4.3); it is read into the Link's factor q_{ij} , the same from both ends. And the Link's pace is the product, $\Gamma - 2c_i - t_a$ to first order [1, every row reads the content; the clock is the Node's, the tension is the Link's]. A positive weight makes a hollow, an attraction; a negative weight makes a hill. The pace is the slowing of the clock: a record at a Node of pace p rotates and moves as it would at Γ , with its intervals p/Γ as long. Every family reads the holders of the content that its declaration names, at the weights it names. Gravity's holder and the binding holder's are read at the weight 1 by every family. The holders are among the readers, and a holder of the content reads its own level among the rest (the law's line, implemented; Section 3.7; the nuclear holders of Section 10.3 are read by the nucleon families alone). So $x = (L_b + L_g)/\Gamma$, with L_b the binding holder's level and L_g gravity's. A family of dimension 2 also reads the holder of the sign plainly. Light, the holder's own record, reads the holders of the content as every family does, and not the holder of the sign. And a family sources what it reads, at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's].

The one assumption of the forms: a Link is clocked twice by its reading Node's own clock. Nothing in Rule3 fixes how a level enters the paces. Beside the line, the model holds that a level is a count of units and that every act is measured in the Node's own units; this is its second input beside the line (ASSUMPTION). Two parts of the paces follow from that. The clock composes, $p_0 = \Gamma(1 - 1/\Gamma)^c$: each unit slows the clock as it stands by the same fraction, since the write of one unit is an act measured in the Node's own units, the read included (DERIVED from it). And every act of the interval, Rule3's step, the write of the held rows and the turn of the sign holder, is Rule3's act measured in the cell's units. So a uniform level $U = c/\Gamma$ changes only the units: time by $N = p_0/\Gamma$ and length by $h = p_0/p_a$ per Link. The write carries $p_x p_y p_z / (p_0 \Gamma^2)$ on the form and $p_x p_y p_z / (p_0^2 \Gamma)$ on the Wronskian, and the turn carries p_0/Γ , per proper volume and per proper interval (DERIVED, S.34; implemented, Section 3.7). One sentence is not in Rule3. It is the model's one assumption of the forms, and it is named as the credit is (Section 5): a Link is clocked twice by its reading Node's own clock, $p_a = p_0^2/\Gamma$, that is $hN = 1$, the ruler going with the clock (ASSUMPTION). The lattice says why twice, since a Link has two ends. It does not say why the reading Node's clock stands at both ends rather than the product of the two ends, which gives $\beta_K = -1$. The sentence is fixed by two clicks, the bending's $\gamma = 1$ and the perihelion's $\beta = 1$, and this is where Einstein's $\gamma = \beta = 1$ enter the model by hand (ASSUMPTION, FITTED: an input chosen with two known clicks in view; every form of Section 8 is conditional on it, and none is a consequence of Rule3 alone; the bending and the perihelion fixed it and are no test of it, while Shapiro's delay, the redshift's factor $f(\omega_0)$ and the shadow's $2em$, e Euler's number, remain its predictions). Under it the model's static metric for light is $ds^2 = -e^{-2U} dt^2 + e^{2U} d\mathbf{x}^2$, to $O(U/\Gamma)$ and in the ray limit. This is the exponential metric [42], Einstein's to the first post-Newtonian order. The clock's rate is N : exactly in the lattice's variable $2\sin(\omega/2)$, and in ω up to the finite-rotation correction $\omega_0^2/12$, which is 5.9 percent at the computed pair (6.3 exactly) and 10^{-20} at nature's LHAASO bound (4×10^{-29} at the axis bound). The metric is stated in the variable in which it is exact (DERIVED; S.31). The shadow's capture radius $2em$ is its second-order click, against Schwarzschild's $3\sqrt{3}m$ (Section 8), in the lattice's frame; that frame is a second residue, not derived from Rule3 and read by the binary pulsars' α_1 (Section 10.3). The quadratic $(\Gamma - c)^2 + c^2$ is the second-order truncation of the composition [1, the same clock and the same ruler for every act; the paces compose][1, what is open, item 15].

The band at a pace. From the coefficients, the rotation of a record at a Node with clock pace p_0 and Link paces p_a is (DERIVED, GAMEBOARD; S.1)

$$\boxed{\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a),} \quad (4)$$

The mass term scales with the clock's square, and the band term with the square of the Link's pace. Two readings follow. First, at $k = 0$ the rest rotation of every massive band obeys $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$, the clocks' shifts of Section 7.4. Second, along a Link the speed of every wave falls with the Link's pace, $\Gamma - 2c - t_a$ to first order, so light is delayed and bent by twice the clock's share. At a Node of content $x = L/\Gamma$, light's band is $[2 - 4N^4, 2]$ in $2\cos \omega$, with $N = (1 - 1/\Gamma)^L$, and at long wavelength $\omega = N^2 |k|/\sqrt{3}$. So light's speed is $N^2 = e^{-2x}$ times its vacuum speed, to the unit: an index $n = e^{2x}$, which is $1/(1 - 2x)$ to the first order, with $n - 1 = 2x$ (S.2). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole. Light meets a total mirror where $p_a < \Gamma \sin(k/2)$ [1, the band at a pace; the rows against nature, (b)].

The guard. A pace is positive by the composition, and it must lie below an edge P , with $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + |\text{num}|)$, which is Γ^2 for light. Beyond P the mode at wave number π on every axis grows without bound; at or below P no mode grows exponentially (THEOREM, GAMEBOARD). The guard is the Courant, Friedrichs and Lewy condition of the explicit scheme that Rule3 is at the vacuum's pace [41], and it is checked once at load. Under the composed paces no Link ever

closes. The clock $\Gamma(1 - 1/\Gamma)^c$ is positive at every content, and the Node's pace is its square over Γ . So a deep well freezes a clock and closes nothing, and the horizon of the quadratic form at $\Gamma/2$ is a feature of the truncation and not of the law (DERIVED, GAMEBOARD; S.3). A Node whose Link pace rounds to 0 is a frozen Node. It reads nothing, and no event enters it. Its level is kept: matter's rotation is nearly stopped, and light's drifts linearly. Its active neighbours still read it at their own pace, so the cut is one-way. Its share is 0 by the NodeReader's declaration, so it is dark (S.3, S.41). Two thresholds carry the word: the Link's pace at $(\Gamma/2) \ln 2\Gamma$ levels, 28,178 at $\Gamma = 6,000$, and the engine's 28,206 with p_0 rounded before squaring (S.41); and the clock's at $\Gamma \ln 2\Gamma$. The guard's bound, $|2 \cos \omega| \leq 2$ at the band's edges, admits the edge itself, where a linearly growing solution exists, and it is stated for $\text{num} \geq 0$. At that edge the vacuum's checkerboard mode, driven by the remainders, grows as $t^{3/2}$; no click reads it, and it reaches the loader's amplitude bound, 9,266 levels on the shipped files (GAMEBOARD). For a negative numerator the lowest rotation lies at $k = 0$ (S.3). At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content is 0 (Section 6.5) [1, the guard; the frozen Node has no share].

3.5 The write, the readings and the interval

The one write. Every write into a holder's level is one act, the second line of Eq. (1). At a Node, in each interval, a held part gains $(\sum_f w_f q_f + r) \text{div } E$: one division per part, with one remainder r carried at that Node. Here q_f is the sourcing family's own quantity at the Node: its form D_f for a holder of the content, its Wronskian W_f for the holder of the sign, or its tension $T_{aa,f}$ for an axis part. w_f is the weight with which that family reads the written holder. E is the part's wall: $E_s T$ for the time part and $E_s W_c$ for an axis part. The sources are summed in the numerator and never divided apart, so there is no remainder per sourcing family [1, the primitives][3, the one algebra].

The readings. A reading is a bilinear form of the levels with declared coefficients. It writes nothing on the GameBoard. The law reads three of them, the third line of Eq. (1). The form $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$ sets the present against the product of the future and the past. For a standing record of peak B it is $B^2 \sin^2 \omega_b$ at every interval. Its quotient over T , the well, sources the holders. The Wronskian W_i is the charge (Section 5.5). The current $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ into the Node i from its neighbour j is antisymmetric. With the Link's factor q_{ij}^2/Γ^2 it is the Link term of the conserved form (Theorem 3); where no tension stands it is the plain current. A wave travelling from i to j has $F_{ij} < 0$. A static record has $\text{now} = \text{next} = \text{before}$, so its form and its current vanish: a static record carries no quanta and no NodeReader sees it [1, the booking; the conserved form].

The interval. Every read takes the values at the interval's start. Rule3's line reads one Link per interval, and nothing in zero time. Every reading of the interval, the step, the tensions and the hold's write, is read from the record as it stands at the interval's start. So a Node's NodeState after one interval depends on the Node and its six neighbours alone: the interval's dependency radius is one Link (the law's line, GAMEBOARD) [1, the interval's dependency radius is one Link]. The interval has four acts, in order: (i) the read, which takes each family's paces from the holders' levels, the reading holder's own among them, and each Link's factor from its tension; (ii) Rule3 on every record; (iii) the readings, which take each record's currents, tension, form and Wronskian at the levels the step started from, and each NodeReader's report; (iv) the write of each holder from its summed sources. [1, the interval][2, the interval].

What is conserved, and where. On the GameBoard the conservations are per family. They are: the weighted form of a family while its reads are static (Theorem 3); the Wronskian of a plane family under a static level, exactly in the line under the implementation's present convention of the turn (Section 5.5); the quanta through the Ports; the wave number of a free record; and reversibility bit for bit. Energy is conserved within a family by theorem, and between a charged body and its light by the law's line (Section 10.1). Between a body and the holders of the content, its conservation is replaced by the mass defect (Section 10.3). The GameBoard conserves no total: the sum of the families' forms is not conserved and not assumed, since the share moves with the paces and no current compensates (Eq. (13)). The click layer has books of its own. A click credits one whole quantum at the NodeReader's own whole, drawn by the inflow of the record's share through its front over its window. A frequency of nature is read as the inflow's beat within the window or as the resonance of the NodeReader's own record. The GameBoard's quantities, the form, the content, the level and the rotation, are read by no click except as the share's inflow. And the credit's bookkeeping is that the shares sum to what the screen saw, one whole per click, where the whole is the NodeReader's own transition (Sections 5.3 and 7.4). In the clicks there are conservations other than the board's, and neither layer keeps a total energy (THEOREM for the board's, ASSUMPTION for the credit's) [1, the conserved form].

3.6 The postulates

The model rests on the law's seven postulates [1, the postulates], with the NodeReader's write added by decision and its gate pending [3, the two hands close the line of the click's write]. (1) The world is Nodes and Events. An Event is a level stepped by Rule3 with its remainder. A click is a NodeReader's report and, by that decision, its one write at the Node the quantum reached it from. Nothing else happens. (2) A Node holds bounded local information, read from itself and its six neighbours, and nothing of the past is kept beyond the one level before. (3) Consistency is local and causal, one Link per interval for every reading of the interval (Section 4.3), with one exception: the credit of a click by the shares over every declared NodeReader, the one non-local element, not derived. (4) The causal speed is one Link per interval. (5) Every physical calculation is on bounded integers, with no floating-point number, no root and no random draw in the GameBoard's step; the NodeReader's draw is its own, with its declared seed. (6) A measurement is a NodeReader's click: the net current into a declared region through its front boundary over one interval, accumulated over the window, and never a Node. (7) The GameBoard, where no one measures, is deterministic and reversible to the bit; a NodeReader's write is the NodeReader's act, forward only and outside the reversal; and the NodeReaders' clicks are the only thing claimed to represent nature. The line and its clock are fixed: every mend of the model is a family, a holder, a declaration or a read, never a change of Rule3 (ASSUMPTION) [3, nothing that is not Rule3 remains].

3.7 The implementation

The implementation, Universe24 [43][2], is the law's implementation and nothing else. It is a GameBoard of Nodes, with every family's NodeState at every Node, and one interval of acts in the law's order. Each act is a call of Rule3 on whole-board arrays of integers at a declared width, every neighbour read through a Port. The call is the same at every Node and for every family; a family's pair, dimension, reads and weights enter it as arguments from the universe file and from nothing else. It holds no number of physics, no formula, no family's name and no flag; every physical value comes from the run's files. A run is headless and writes the click lines, the measurement, and it runs for results only after the check of Section 4.1 says MATCH [2, the engine; the NodeState; the output; the gates]. It holds the law's lines of Section 3.4, and under them the checked files' outputs are bit-identical to the quadratic forms' runs [2, the NodeState; the interval]. The checks' runs are these. The two slits: the Huygens prediction 273, written before the run; the run $N = 278$, five quanta over the blind, 0.3 of $\sqrt{273} = 16.5$; and the law's real line 277.8 (278.0 in the physical labels), computed after the run by the script `two_slits_real_line.py` beside the paper. Bell: $S = 478/169$ with the marginals $1/2$. GHZ: the eight shares, every pairwise E 0 and $M = -4$. And the back-in-time check: MATCH on every file (COMPUTED, runs of the implementation; CLICKS) [1, the rows against nature, (g) and (h); the GHZ gate]. The paper reports the checks' runs and derives the rest: a run of a declared file computes what the algebra of that file states, and the implementation is held to the law's lines by its tests and the gate. The implementation is public, with the documents, the files and the scripts, in the Git repository <https://github.com/Closer24/Universe24>, archived at Zenodo [43]. It installs in a clone with `python -m pip install -e ..`. The two slits' file runs with `PYTHONPATH=src python tools/run_inputs.py --out runs/first_examples/events/two_slits/two_slits.json` and is read with `tools/click_counts.py`; Bell's and the GHZ's files run with `tools/bell_gate.py`; the back-in-time gate runs with `tools/back_in_time.py`; each as the engine's document states it [2, how to run a world]. The paper's own scripts stand in `paper/general_formula/`. Every act of the engine is derived in the supplement from the law's lines, or is named there as declared, a named assumption or open, in one ledger of the two derivations (S.61), with no claim of nature.

The rules we took. What this paper computes is one chosen set of rules: Rule3 with the one write and the three readings (Eq. (1)); the postulates of Section 3.6; the primitives the files declare, each under its own name (the families with their pairs and dimensions, the holders with their sources and divisors, the NodeReaders with their four declarations, the faces); and the integers of `rule.json`, Γ , T , the width and the Link unit (Section 6.6). The engine fixes Rule3's form, the recurrence, and nothing else of the set. Another universe file, other families and primitives and other integers can be taken, and other things built with it [2, the interval; how to add a family]. Everything the paper derives is derived from this set, and it claims nothing beyond it. Where a derived form coincides with a known one, the text says derived and names the form it meets; it never presents the known form as the law's own.

The computation is elementary. The computation reported here is elementary, and it is offered so that others may carry it further. The GameBoards run are small: 25^3 and 41^3 Nodes, planes of up to 45×48 and chains of up to 128 Nodes. A body spans a few Links at about 10^3 levels per Node. The integers are kept to 63 bits. The step is a Python loop over whole-board arrays, and no symmetry of the lattice, no derived resolution and no fixed-point lay is used yet. The lattice's pixel is therefore coarse in every computed body, and its budget, $(\text{Link}/R)^2$ and $\sqrt{n}/A$, is of the order of percent, stated against nature's pair in Section 7.3. What this paper derives, it derives by the law's lines. What it computes, it computes on this elementary engine, and marks as such. The gates of the bodies and the atom are named open: designed and not claimed, with the click's write built after the freeze (Sections 7.3 and 10.3). The method's room for improvement is the paper's invitation.

4 Exact properties

This section states what holds exactly on every lattice, each item proved in S.48: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record (S.5, S.6).

4.1 Reversibility

Let σ be the direction, +1 forward and -1 backward; let div and mod be the floor division and its remainder in $[0, w)$; and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_j R_{ij} \text{arr}_j + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and the backward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

(Proved in S.48.) Theorem 1 holds for the GameBoard's step at every interval. A NodeReader's write and a body's window, click or not (Section 5.4), are the NodeReaders' acts outside the reversal, as the lay is. The back-in-time gate is exact between clicks, and runs across a click and every erasure interval with the presented face values as its boundary. Those face values are read from the GameBoard-labelled lines beside the click line, with the lay's acts crossed from their lines; the click line itself names no Node (THEOREM for the step; the write ASSUMPTION). The whole run therefore goes back: N intervals forward and N back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM where the writes change no level, by construction from Theorem 1).

The implementation checks this as a test: it runs a file forward and back and compares every array at every interval. On the files of Section 9 it reports MATCH over their whole runs; these are files whose write stays under one level (Section 6.6) and whose NodeReaders write nothing. The interval's reversal with a level-changing write is gated. The back-in-time check gives MATCH across the faces, the NodeReaders' clicks and every erasure interval, with the presented face values as the boundary, and MISS at a lay, the taking, the giving and the null window, by design, since the tool's crossing of a lay from its lines is not yet built (Theorem 1's clause), on the three files (the two slits 128, Bell 98, the GHZ 58 intervals) (COMPUTED, GAMEBOARD) [1, the inverse; the objects][2, how to run a world][3, the Node knows no family]. The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the step back. So a record's rotation sense is never read at a Node, and the holder of the sign is read by the record's shape, a plane or a line (Section 5.5) [1, the interval][3, the board goes back in time, as a gate; there is no sense].

4.2 What the acts can form

Rule3's line is one act, and declared coefficients make four acts of it: (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

(Proved in S.48.) So every product of two levels is a reading and never a write, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests. It must be generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, with nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root. The integer square root appears only when a file is laid, as the fixed point of the division act $x \leftarrow (x + n \text{ div } x) \text{ div } 2$ [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

4.3 The conserved form and the momentum flux

The read coefficient $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$ carries the Node's factor and the Link's. Where the paces differ, the Node's factor is not symmetric and the Link's is. So the conserved form carries the weight $1/p_i^2$ on its Node terms and the Link's factor q_{ij}^2 / Γ^2 on its Link terms:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i}{p_i^2} - 2 \text{ num } \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{ now}_i \text{ before}_j. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces for which a positive Node weight d_i with $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$ on every Link exists, E with the weights d_i is exact under the line without the rounding. Under the law's read coefficient, the Node's pace squared times the Link's factor squared, $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$ with $q_{ij} = q_{ji}$ and $p_i = p_0(c_i)^2 / \Gamma$ the composed Node factor, the weight $d_i = 1/p_i^2$ cancels the Node's factor on both sides of every Link. So E is exact for any static content and static tensions. Its Link term carries the Link's factor, so the current of Rule3's own conservation law through the Link ij is $\text{num } (q_{ij}^2 / \Gamma^2) (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$, which is the plain current F_{ij} where no tension stands, $q_{ij} = \Gamma$. The form with the plain current holds there alone; with a tension it is not conserved (two witnesses, S.32).*

(Proved in S.48.) With the rounding, each Node's remainder adds the term $(r - r')(\text{next} - \text{before})/p_i^2$ to the walk of E (DERIVED; S.5). E is a diagnostic. Where the paces move, the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ changes exactly by $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$. This change is 0 at fixed uniform paces, and negative where a well deepens under a standing record (THEOREM, by substitution; GAMEBOARD) [1, the conserved form].

The momentum flux is the wave's own stress. The momentum density along the axis a at the Node i is the current per axis, $P_a(i) = (F_{i,i-a} - F_{i,i+a})/\text{num}$. At constant paces, the line without its division gives the identity below exactly. The integer step adds the remainder term $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$, with $\delta = r - r'$ and Δ_a the centred difference, which the implementation's test includes (THEOREM for the line, GAMEBOARD),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i) - G_{ab}(i-b)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with $G_{ab}(i)$ the stress reading on the b -Link ($i, i+b$), from the levels now alone. The proof is in S.48. The tension on the axis a at a Node is the part a holder reads into the Link's pace, and the source of a holder's axis part. It is the momentum flux averaged over the Node's two a -Links,

$$T_{aa}(i) = -\text{num } (G_{aa}(i) + G_{aa}(i-a)) \text{ div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

and each Node sources its own part, $-\text{num } h_a(i)$ with $h_a(i) = \text{now}_{i-a} \text{ now}_{i+a} - \text{now}_i^2$, read from its two neighbours along the axis at the interval's start. The Link's tension is the parts of its two ends: one number per Link, read the same from

both ends by the pace at the next interval. A plane wave of amplitude b at the wave number k has $G_{ab} = -2b^2 \sin k_a \sin k_b$, so the flux along the wave, $T_{aa} = -\text{num } G_{aa} \geq 0$, is positive, as a wave's pressure is (S.7). A packet of count c and velocity $\mathbf{v}$ has $2c v_a v_b$, the two Links' currents summed, which is the stress of dust (S.7 for the plane wave). With the sign as written, a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the record's own stress].

4.4 The count is the record's share

A family's count of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = \left[w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_{j \sim i} R_{ij} \text{before}_j \right] \text{div } (2p_i^2), \quad (9)$$

the conserved form's term at the Node with its weights (Theorem 3), the Node's $1/p_i^2$ and each Link's q_{ij}^2/Γ^2 inside R_{ij} , read in quanta over W_c as the nearest whole number. At the vacuum's paces it is exactly 3 den $(\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before})_i$, with S_6 the six-neighbour sum. Read with the vacuum's coefficients instead of the Node's own paces, the same expression is the vacuum's share, the unit of the tables of Section 7. The implementation reads it so. Over one step of Rule3 the share's rational value, before its floor, changes by exactly the six currents into the Node, each weighted by its Link's factor, plus Rule3's own rounding (THEOREM for the rational identity, GAMEBOARD; S.5). The integer share of Eq. (9) adds the reader's own floor, a residual strictly between -1 and 1 per Node and level pair, distinct from Rule3's remainder:

$$e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) = \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} F_{ij} - \frac{(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2p_i^2}, \quad (10)$$

with F_{ij} at the levels the step started from, and r, r' Rule3's remainder before and after the step. So over any region the total share changes only by the weighted currents through the region's boundary, the remainder terms and the readers' floors (THEOREM, GAMEBOARD) [1, the count is the record's share; the share's change is the currents]. A formulation that moved whole quanta by a division of the current would put numbers at every Node that no act of Rule3 reads back. The law does not carry it (ASSUMPTION) [3, the count's line and the sense's line leave the law]. Where the paces move, the share drifts with no current crossing any Link, Eq. (13); this is a diagnostic [1, the paces' own change moves the share with no current].

The signed integers. The levels are signed integers. The currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of a closed or periodic GameBoard is non-negative for $|\text{num}| \leq \text{den}$, with equality for a static uniform level of light and, on an even periodic board, for its checkerboard (THEOREM, GAMEBOARD; S.6). One Node's share can be negative at a lay's edge, and a region's by at most half a level-square per boundary Port. Over the whole passage of one forward packet, the inflow accumulated at a region's front equals the region's final share plus what left through its back, and each of these is bounded below by S.6 (Eq. (10) summed over the window). So the inflow is non-negative up to the rounding, and the credit by the shares is well defined. Where a window is cut inside a passage, the NodeReader credits $\max(s_R, 0)$ (DERIVED, CLICKS) [3, the law's integers are signed].

5 Measurement: the click as a meeting

The click is where the method meets Universe24, the clicks it computes. The click is the meeting of the future with the past at a declared NodeReader; it is no act of the GameBoard. A click is a fact: one whole quantum passed from a giver to a taker, written once and never undone. It is always the meeting of two quanta, so every click has its giver and its taker, the arriving record and the transition of the record that waits for it, and never happens by itself (S.49 (16)). In the law's one line, a NodeReader drives Rule3 from outside the board, as a face and a start do. It presents one value at one Port of the click's Node, in the place of what the Node would see from the board, and lays whole quanta there; the Node steps Rule3 as always and knows neither. A click is this act: one whole quantum of one record is taken, Rule3 writes the hole, and whole quanta of others are given, the rotations summing at the Node. The conversion, the weak force, is the same act with the family changed on the out side; every other click is of its kind with the family kept. The draw alone chooses whether and where, which future continues. In the words of the division, a click is Rule3's division act on the families. The shares that entered a NodeReader over its window, with the remainder the books carry, are divided by one quantum W_c ; the quotient is the clicks and the remainder is kept. What Rule3 does to the levels at a Node with the wall w , the credit does to the shares at a NodeReader with the wall W_c , with the draw in the place of the Port (the three storeys of the one act, S.48). The engine writes a NodeState by Rule3, by the lay and by the face, and by nothing else; it writes the hole by the face value at the click's Port (the implementation of 2026-10-03; the three gate files' numbers hold bit for bit under it). The credit's window it rounds half up and drops its remainder. The carried remainder is the law's form, stated under its own name with its own blind on N ; it is the top storey's "how many" and never its "where". The board's remainders cannot stand where the draw stands: a local carry per NodeReader gives no shot noise, and Rule3's remainders used as pick numbers put every click of the two slits on two of twelve regions (a recorded check; Section 10.3). This section states what is on the lattice, the meeting, the NodeReader, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

5.1 Events, and no waves

On the GameBoard there are events and amplitudes. An event is a level split by Rule3 to the six neighbours. An amplitude is the level Rule3 produces at a Node, the sum of the split events there. The record is the amplitudes over the GameBoard.

In Universe24 there are clicks, the NodeReaders' reports, and nothing else. In the two slits the lay's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events. Read backward from the realised click by the same line, Rule3 at -1 , they form the past's tree, which also passes both gaps back to the root. What stays on the GameBoard, the amplitudes no click realised, is not in Universe24. After a click they are a cancelled future at three levels. In the credit, the record's count is at 0 and no NodeReader credits it again. At every NodeReader they reach where the file declares a face behind it, their share leaves the board there. On the board, the record that gave is erased from the click's Node outward at the causal bound, one Link distance per interval, inside the click's light cone; beyond the front its levels remain, an empty wave that still sources the holders it writes to, until the front arrives (Sections 5.7, 5.4) (ASSUMPTION) [1, no waves on the event board][3, no waves on the event board].

5.2 The click is the meeting of the future with the past

Assumption 1 (the click is a meeting, the law and not a hypothesis). A record runs on the GameBoard in both directions by the one line: forward from the root, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at -1 . A click is the meeting of two quanta at a declared NodeReader: the quantum of the incoming record run forward, the future, and the transition of the record that waits for it, the click's unit, the past read back from it. Its share is their product at the region, at resonance (Section 7.4): the form $D = \text{now}^2 - \text{next} \times \text{before}$, the now against the future times the past. Which meeting is realised is the NodeReader's draw with its declared seed, and the GameBoard draws nothing. The one non-local place is the credit's one draw over the NodeReaders that read one record, through the root. The NodeReader only asks whether the future and the past met at it.

There is one kind of reader, the NodeReader, and the file decides what it holds. A NodeReader with no record of its own reads the record that arrives at it, a bolometer's front; every check reported here ran on such NodeReaders (Section 5.3). A NodeReader whose file gives it a record of its own, with parts, transitions and a lifetime, stands for an atom or an ion of nature: it reads the arriving record against its own transition, and it gives as well (Section 5.4).

The click, algebraically. Let X be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step S_t is Rule3 over the interval, with the paces read from the holders at its start; it is a bijection of X (Theorem 1). Its forward orbit is the future and its inverse orbit is the past. A NodeReader R is a declared region with its front Ports and its coefficients, which define Q_R , the inflow through its front over the window (S.50). A click is the pair (R, t) : the credit of one whole quantum to R at the interval t , drawn with the probability $s_R / \sum_{R'} s_{R'}$, where $s_R = Q_R(x_t) / W_c$. The decision is the one map in the model that is not a bijection. It is a scalar under the group, so no click names a Node to us. Its write is one act on X , at the one Node of the region that the draw picks by the share (Section 5.4): the levels that arrived there are taken, the record's count falls by one and the taker's count rises by one, the quantum is conserved between them, and the change spreads by Rule3 as an event (Section 5.4; ASSUMPTION, the NodeReader's declaration). A frozen Node stands in no NodeReader's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS) [1, the click's algebraic definition].

One form for every click. At the credit the parts of a record are read with the NodeReader's coefficients, each part paired with the same part through the common origin, then summed and squared. The share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's check is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). In the implementation the joint share is computed from the forward records alone, by the linearity of the rule; the check of Section 4.1 shows that the backward reading exists to the bit (the credit ASSUMPTION; the equality of the forward and backward descriptions DERIVED for the line at fixed paces; CLICKS).

5.3 The NodeReader, the one declaration kind

Assumption 2 (the NodeReader's declaration). A NodeReader is one connected region of two Nodes or more declared in the file, with its seed, and never one Node. Where it reads a travelling wave it is at least half the wavelength of the family it reports across the beam (4 Links at $\lambda = 8$); a narrower region, a single Node or a region in pieces is refused at load. It is opaque, backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns. Each interval it reads the inflow through its Nodes' front Ports, the Ports leading in from the declared board, Rule3's currents summed in integers into its window. Its click is its report, one line per quantum: the window's index, the NodeReader's name, the family and the part, the count before and after, and the NodeReader's own clock, never a Node. One wall $W_c = 3 \text{ den } T$ of the inflow over a passage is one unit of count, the bolometer's whole ($W_d = T$; a counter's whole is $T \sin \omega$, S.49 (14)), so a screen saw N units where the inflows of its regions sum to $N W_c$. The credit is one quantum to one NodeReader, drawn by the shares of the inflows of the regions, read outside the GameBoard over every declared NodeReader of the file; one credit per quantum and never two. The NodeReader draws with its declared seed, the GameBoard draws nothing, and the run is deterministic per seed. After a realised click the NodeReader writes once, at the one Node of its region that the same draw picks by the inflow booked through each Node (Section 5.4). A NodeReader whose file gives it a record of its own lays that record over its Nodes in the counts' proportion, with its parts, its transitions and its lifetime, its outer Links cut and its inner Links open, so that its uniform mode rotates at $\cos \omega_0 = \text{num}/\text{den}$ exactly whatever the region's size; it is held by its declaration and never bound by a well, and it gives as well as takes. The credit is the one non-local place of the description and no line of the law.

What produces a click is deterministic: the meeting. Which of the possibilities becomes the click, no line of the law gives; the NodeReader draws. The draw decides a place and a combination, never a value and never a path. How many clicks is no draw: a NodeReader reading the record credits at the window's close every whole quantum its inflow holds, N the sum of the shares over W_c , capped by the record's count, each quantum with its own draw of the region and of the

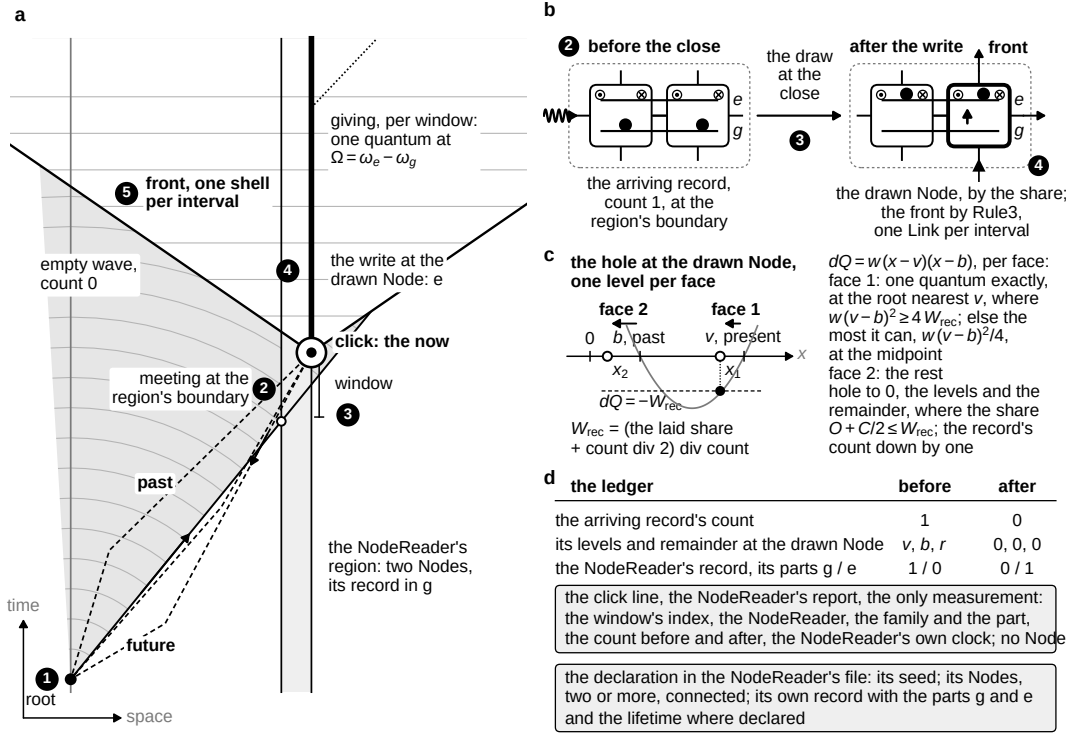


Fig. 4 The click at a NodeReader, in plain words (ASSUMPTION for the draw and the write, the rest DERIVED; CLICKS; drawn from the definitions, no run; solid lines run forward, dashed lines read backward). A NodeReader is where the future meets the past: the arriving record's quantum, run forward from its root, is the future; the quantum of the transition that waits for it, read back from the realised click, is the past; the click is their meeting, the present, and there is no click of one quantum alone. (a) Space and time. 1: the arriving record runs forward from its root by Rule3, every path at once (the grey fan). 2: it meets the NodeReader's region, two adjacent Nodes (the grey strip), at the region's boundary (the open circle); the window (the bracket) opens at the NodeReader's last write and closes at its draw. 3: at the close the NodeReader draws with its declared seed by the shares of its Nodes, once over every NodeReader that reads the record, through the root: the click, the now (the ring). From the click the same line is read backward, Rule3 at -1 (the dashed lines): a reading, never a write. 4: forward, the one write, at the Node the draw picked by the share (the thick world line); the NodeReader's record goes to its part e . 5: the change spreads by Rule3 at one Link per interval, the front (the white cone), and erases the record that gave inside the click's light cone, the ball of $2d + 1$ zeros at the distance d , never faster (Section 5.7); beyond the front its levels remain, the empty wave with count 0, stepped on and never credited. The dotted line is the giving when it comes: one whole quantum of light laid at that Node at $\Omega = \omega_e - \omega_g$, drawn per window at the declared rate plus the floor. (b) The region: two Nodes with their Ports, the dotted boundary the declaration's; the NodeReader's own record is laid over both with equal levels, its parts g and e the ladder, so that its uniform mode rotates at $\cos \omega_0 = \text{num/den}$ whatever the region's size. Before the close (2) the record is in g ; the draw (3) picks the drawn Node by the share; after the write (4) the drawn Node (the thick border) carries the record in e , and the front leaves through its Ports. (c) The hole at the drawn Node, by the booking identity $\Delta Q = w(x-v)(x-b)$, x the written level, v the present and b the past, each face writing one level: face 1 takes one quantum exactly at the root nearest v (x_1) where $w(v-b)^2 \geq 4W_{rec}$, else the most it can, $w(v-b)^2/4$, at the midpoint; face 2 takes the rest (x_2); where the booked share $O + C/2$ is at most W_{rec} the hole goes to 0 and the count falls by one; $W_{rec} = (\text{the laid share} + \text{count div } 2) \text{ div count}$; no factor scales the hole. (d) The ledger in whole numbers, the click line (the window's index, the NodeReader, the family and the part, the count before and after, the NodeReader's own clock, no Node: the report and the only measurement) and the file's declaration (the seed, the Nodes, two or more and connected, the record with its parts and lifetime where declared).

Node and its own click line, and no cap of one click at a time is added; a NodeReader with a record of several parts turns and draws its outcome once per window. Which region is drawn by the shares; which boundary Node, by the per-Node inflow; which combination of the sides' ports, through the root. The generator is a declared linear congruence with its seed, of the full period 2^{width} by Hull and Dobell's theorem for every shipped generator. So Born's rule on the board is the proportionality to whole shares, exact over the period, and nothing of the Node and nothing of Rule3 enters it. Exclusivity, one click per quantum, is derived under the credit's one draw, the NodeReader's declaration, from the count conserved (Section 5.4). Born's rule, the credit's chance equal to the intensity's share [44], is a declaration of the NodeReader and not a finding of the model [1, what produces a click is not known]. The uncertainty relation is the NodeReader's three declarations and no rule of its own: the region, $\Delta y k \geq \pi$ at half a wavelength; the window, $\Delta \omega \geq 2\pi/N$ over N intervals; and the draw, $\sqrt{Np(1-p)}$, with a floor and a whole. The write is exact inside, at a Node and an interval that are the board's alone and never reported, and the report is uncertain outside; the two are one act. The NodeReader is defined by the uncertainty relation, and that is why it stands on two Nodes or more and never on one: one Node has no wave number to resolve (ASSUMPTION). A NodeReader is a face of the GameBoard that reports: the Nodes see a boundary, the world sees a click. It simulates the click's effect and not its cause. In nature what clicks is bodies, and what it does happens in those bodies; our NodeReader is not in the Nodes. Its acts are the loop's from outside the Node, as the lay and the receding face

are. The share that reaches a region backed by a receding face, as every gate file's is, leaves the board there, credited or not; that is a declaration of the file and not a rule of the NodeReader [3, the uncertainty relation is not a rule of its own]. What each declaration costs is derived in S.8 and S.50: a pattern in the histogram and never in one click (DERIVED, CLICKS) [2, the output].

The NodeReader is one declaration kind. The NodeReader is one declaration kind for every experiment: its seed and its Nodes, two or more, and, where the file gives it, a record of its own, and nothing else. Its window is declared and is never read off the wave: the file's window for a NodeReader with Nodes alone (the shipped gates declare it their run), and the tick of the probe's lay, or the declared window where no probe stands, for a NodeReader with its own record; the declaration is set, explicitly, to cover the support of the arriving share at its Nodes, both levels, for a finite packet and for a record of its own alike, and for a continuous flow the cut is declared and stated as such. Its unit for an arriving record is the record's own quantum $W_{\text{rec}} = (\text{the laid share} + \text{count} \div 2) \div \text{count}$, read once from the books' origin, or at the first lay for a record born later, and $W_c \sin \Omega$ for a born quantum, so that no transition is declared on a region; for its own record the unit is that record's transition: one click is one quantum's change in the absorbing record, $T \sin \Omega$ at the frequency of the NodeReader's own record, never the event's division by its own $\sin \omega$ and never a change of the share. For its own record the arriving level is read into each part's phase at the declared weight and resonance pair [num, den], $\cos \Omega = \text{num}/\text{den}$: the part's two time levels turn by Rule3's rotation act, and the labels' squares at the region's Nodes after the turn are the shares of the outcomes. It draws at the window's close by those shares; in the dark it draws at its clock's grain, over the grain's span of board intervals or the lifetime where that is shorter, with the weights [$\text{span} \times p_0 \times \text{unit} \div \Gamma$, half up; $\tau \times \text{unit} - \text{that}$], the hazard span/τ in its proper time, so that the exponential is read in proper intervals. It gives the quantum at $\Omega = \omega_e - \omega_g$, read from its modes. Its own clock, the carried sum of p_0 over Γ at its Nodes, and its event clock, the windows closed, stand in the click line beside the board's tick; the click line is the experiment's whole reading, and every GameBoard number, the Node written, the levels taken, the erasure shells, the regions' densities, is a diagnostic labelled so (DERIVED, two derivations; CLICKS; the implementation's versions and their readings are in S.62; S.61, lines 12, 13 and 17).

5.4 The click decides in the credit and writes at one Node

Assumption 3 (the click's decision and its write). The GameBoard is deterministic and local: Rule3 with no draw, and the bijection of Theorem 1 at every interval. The click is the meeting of two quanta at the NodeReader: the incoming record's quantum and the transition of the record that waits for it. The NodeReader, declared in the Universe24 file, draws with its declared seed whether the meeting is realised; the randomness is the NodeReader's, and the board has none. Forward, to the future, one write: the exchange at the one Node of the region that the draw picks. The NodeReader knows that Node and we do not (the uncertainty principle: for us the click is a region and a window). That one Node is drawn in the same one draw, by the window's inflows per Node, whose sum is the region's share; for a NodeReader giving from its own record, by that record's share over its Nodes; the Node is a GameBoard diagnostic beside the click line, as the implementation holds it. The exchange: one whole quantum passes between the two records at that Node, the one generic act, a face presented at the Node's Ports. The arriving record gives one quantum there and its count in the books falls by one; the taker's part rises by one, laid in its mode with the count's own scaling, $A^2 = (N + 1) T / (2 \sin \Omega)$ on each of the part's two lines, with the phase of the product and the remainder at the origin of the lay; the part it leaves is laid with its count down by one. The hole in the arriving record is written by the booking identity on each face, $\Delta Q = w(x - v)(x - b)$, x the written level, v the present level and b the past: the first face takes one quantum exactly, at the root nearest v , where $w(v - b)^2 \geq 4W_{\text{rec}}$, and else the most it can, $w(v - b)^2/4$ at the midpoint; the second face takes the rest; where the Node's booked share $O + C/2$ is at most W_{rec} the hole goes to 0 at that Node, its levels and its remainder, and the record's count falls by one. A long-wave massless record keeps part of the taken quantum's form, stated as such. Nothing is scaled by a factor and no number enters: the faces write one level each by the identity, in Rule3's integers. The quantum is conserved between the two records, and the taker's change spreads by Rule3 as an event, one Link per interval. No act is faster than one Link per interval: the front erases the record that gave, from that Node outward at the causal bound, one shell of Link distance per interval, the hole's act on each shell, the ball of $2d + 1$ zeros at the distance d ; the clicked record alone is written and nothing else. The other paths are a cancelled future at three levels, and the front erases them inside the click's light cone (Section 5.1). Backward, to the past, a reading and never a write: the credit's reading through the root, which chooses which tree is the past and gives Bell at a distance (Section 9.3). It is the one non-local place, and it cannot be a write, since the board is deterministic forward.

After its click, a single quantum's count stands at 0. The count is conserved and is read by the credit through the root, the NodeReaders' shared other side on the GameBoard; the Nodes' levels carry the shares and never the count. This gives antibunching, and one quantum, one click per record between far NodeReaders [45], from the credit's one draw over all the NodeReaders that read one record (ASSUMPTION for the one draw, DERIVED from it for the rest; CLICKS; the numbers of Sections 9.2 and 9.4 are unchanged, one click per record). No signal passes: nothing at one NodeReader changes the amplitudes at another except by Rule3 from the Node of the exchange. A NodeReader is a body of the GameBoard with two sides, and it is entangled on both. One is the front, where the incoming quantum meets the transition it reads. The other is the side where its record connects to the apparatus's bodies [1, the observer is a cluster], and, through the record it took, to that record's root. So two NodeReaders reading one record are connected through the root, their common past. A bound body clicks when quanta from outside meet its own transition, never by clicks from outside; the meeting is the resonance of Section 7.4 (DERIVED, CLICKS), and it clicks only where it is declared a NodeReader. A bound body has no window of its own, and the now reaches it only through a NodeReader's click; its own period as its window is killed by nature's $n = 1$ (DERIVED, CLICKS; S.59) [1, what is open, item 46]. The window of a NodeReader's own record is the message's duration, as the law can read it; on the integer board a laid packet carries a precursor and a ringing tail that turn, so the support of the arrival closes no window under a laid wave, and no window closes on the wave itself. The window is therefore declared and bounded by the schedule of the lays: it opens at the record's last write and closes at the tick of the probe's lay, the probe's taking in a measurement, and the probe's arrival is derived from its lay, from click times and from no level read. The declared window of a shipped file is the experiment's protocol, a declaration of the file's lays and not of

the NodeReader, which receives its seed and its Nodes alone. Where no probe is laid, the declared window stands, explicitly named, as the probe. The arrival-window rule as built is derived by two derivations, built and read, and enters no shipped file (DERIVED, two derivations; CLICKS). A body with no declaration does the Nodes' laws alone: the two-mode transfer and the radiation at its beat by the hold's write, continuous, with no count moving and no jump. A NodeReader with a record of its own draws over its parts at each window and writes its outcome at the one Node of its region that the draw picks, re-laid in the drawn part at its count there and nowhere else, click or not. A NodeReader reading a light record writes nothing on a null outcome (S.59). A spread body's click is the exchange at its entry Node, and its settling is a reading, named open (Section 10.3). Of the past the click erases nothing on the board: Rule3 between clicks stays the bijection, and the back-in-time gate across a click restores the taken share from the output's click line, the click being the one act outside the reversal. Of the future: the record's remaining share spreads on as an empty wave, a wave with no quantum, interfering and never crediting, and the front carries the record that gave. What is sent on: the changed record at the drawn Node steps by Rule3 from the next interval, so a later NodeReader reads the changed board and the futures that cannot follow do not click. The giving is the lowering, at a Node whose record is in its upper part, drawn per window at the declared rate plus the floor: one whole quantum of light is laid at that Node at the resonance $\omega_L = \omega_e - \omega_g$, read from the modes, its total normalised to the invariant, $T/\sin\Omega$ on the one line laid (the source in time keeping $T/(2\sin\Omega)$ per line), the direction an assumption stated as such; the conversion lays the table's rows at the one Node. The rotations sum exactly at the Node, and the shares sum to the booked remainder, $T(\sin\omega_e - \sin\omega_g)$ against $T\sin\omega_L$, a finite-rotation correction booked explicitly (S.14, S.43). The conservation laws of a click are the write's own construction and no rule added: the count whole, the rotations summing, the charge's unit $T/2$ of the Wronskian per quantum (Section 10.1), every line of a record together. The NodeReader's one act of its own is the draw; the write that follows is derived. What clicks is this. Light clicks with a body's transition (the taking and the giving). A quantum of matter clicks with a body's transition (the capture and its reverse). Light meets light by summing in one row and never by a click. The holders of the content, gravity and the binding row never click (S.49; the conversion of a quantum between families is the hypothesis below). A body in the dark is only the future and still gravitates. Emission is the same meeting with the roles reversed: two local writes and one draw of the credit binding them. The Node stays generic: it knows no family, no NodeReader and no click; the engine's loop performs the NodeReader's acts at the Node's Ports, as the lay and the face do, and Rule3 is never changed. The write is the NodeReader's act (ASSUMPTION, the NodeReader's declaration [1, the click writes on the GameBoard]; the form of the decision is DERIVED, the one form of Section 5.2). The GameBoard holds only the future, every possibility, and has no arrow: Rule3 is a bijection at every interval (Theorem 1). The clicks have the arrow. Each is forward only, the one act outside the reversal. At each click what was future becomes past, and the present is the meeting. The chain of clicks is the past advancing into the future, written once and never undone. The direction of time belongs to the clicks and not to the board (THEOREM for the step; the clicks' arrow is the NodeReader's declaration). The implementation's checks of the click's write are recorded in S.62: the back-in-time check across the faces, the clicks and every erasure interval, the hole's share and the count's fall asserted exactly, and the numbers of the three gate files bit for bit under the face. The checks reported in this paper were run with NodeReaders that report and write nothing, and the write at the window's end leaves their numbers unchanged (Section 3.7) [3, the two hands close the line of the click's write; the detector only writes, and it writes only to the one Node from which it received the click; on the GameBoard there is no draw; the one who makes this draw is the detector; there is no family of clicks at the Node][1, the click is the meeting of the future with the past; clicks go only forward; the backward run is exact].

Only a click moves a quantum between families: a derived statement, and a hypothesis under its own name. Two statements stand under this heading, and each has its own mark. The first is derived, and the implementation holds it. On the GameBoard a record's levels change by Rule3 in its family's own plane and by the holders' reads alone, and the read moves no form; the turn is a gauge. Form passes between a family and a holder's record only through the write at a click: the giving's beat written to the holder's line, and the hole's levels taken at the one Node. A count passes between families only at a click: at a taking the arriving record's count goes down by one and the body's part up by one at the Node; at a giving the part goes down by one and one whole quantum of light, a record with count 1, is laid at the same Node; at a conversion the table's rows are laid at the one Node. Between clicks every count is constant. In one sentence: a quantum changes family or part only at a click's write, and between clicks every count is constant. This is a statement about quanta and the records' lines, not about form or a holder's content: a holder's own content is written every interval from the sources' bookings, and a change of the paces moves a record's form with no current, the mass defect (Section 10.3). The implementation holds it structurally and tests it: nothing writes a record's lines but Rule3, the lay and the face, and the credit's counts move in the click's acts alone (DERIVED, two derivations; CLICKS). The second is a reading of nature, a hypothesis under its own name, on which no number of this paper rests. The GameBoard computes the possibilities of the future locally and carries shares. The clicks are the present, where under the hypothesis the world's real computation is done, each between the past, the body's own transition read back to the record's root, and the future, the incoming record's paths. What passes between families on the board is form, a possibility; what passes at a click is a quantum, a fact. What tests it: a body laid in its excited part alone, with no beat, gives no light by the write alone, and under the hypothesis it gives one quantum at an exponential time, a single atom's lifetime. Its gate files are named in the open list (HYPOTHESIS [1, only a click moves a quantum between families]; Section 10.3).

5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder. A holder's parts stay real. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (11)$$

the Wronskian of the two pairs. The line without its division conserves it exactly, with the weights of the conserved form, under the condition of Theorem 3 on the paces (THEOREM, since **DM** is then symmetric; **GAMEBOARD**; S.9). The integer

step adds to each part the remainder term $\epsilon = (r - r')/w$, $|\epsilon| < 1$. So the integer W changes at a Node by $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$ per interval, not zero in general; one witness is computed in S.9 (COMPUTED, GAMEBOARD). An exactly conserved integer charge is open (Section 10.3). Its local change is the Link current: $(2 \text{num}/w) (q_{ij}^2/\Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$ per Port for the weighted W_i/p_i^2 , and $\text{num}/(3 \text{den})$ times the bracket for the integer W_i at the vacuum's paces (S.9). Its total over the GameBoard is invariant under any change of the paces that keeps them uniform over the GameBoard, and the weighted total $\sum_i W_i/p_i^2$ is exact at any paces, since the bracket $\mathbf{im}_{\text{now}}^\top \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{now}}^\top \mathbf{M} \mathbf{im}_{\text{now}}$ vanishes for every symmetric $\mathbf{M}$ (THEOREM, GAMEBOARD). A real record ($\text{im} = 0$) has $W = 0$, so light, the real record of the holder of the sign, is neutral. A rotating record has W of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by no name (ASSUMPTION, GAMEBOARD): a family of dimension 2 has a Wronskian and reads it; a family of dimension 1 has $W = 0$ and does not. The holder of the sign does not read its own level, so light does not read it and a static charge does not bend light. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1). So the holder of the sign acts on its readers by the act its row declares, **rotation**. The holder's level over Γ is the angle θ that turns the two-part record in its own plane, $e^{-i\theta} z_{\text{next}} + e^{i\theta} z_{\text{before}} = (S z_{\text{now}} + \sum_a R_a (z_{+a} + z_{-a}))/w$. These are three exact integer shears, inverted bit for bit, with θ read at now alone; under them like and unlike senses are told apart by the record's own sense. Which sense moves apart is one bit the model does not fix; it is declared by nature's measurement that like charges repel. The exact angle of the shears is $2 \arctan(L/2\Gamma)$, which is L/Γ at small angles, and the magnetic force follows as the Link's phase, Peierls' coupling (Section 8) (ASSUMPTION for the declared bit, GAMEBOARD) [3, there is no sense; every family has a dimension; the rotation round, built][1, the sign holder rotates the two-part record; the rows against nature, (f)].

5.6 Light is born by the write

Light is the sign holder's own record: one record whose time part is a real level pair. It is stepped by Rule3 at the paces of its reads, so that light is bent by the content, and it is written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing or accelerating body writes a varying source at its Nodes; the sign holder carries it away at one Link per interval, and those spreading events are the light. A uniformly moving body writes no travelling light in the long-wavelength band (S.42 (b)) (DERIVED, CLICKS). A NodeReader reads light's current through its boundary in quanta of W_c . The travelling events of the other holders enter a NodeReader the same way; in gravity's case they stand beside nature's gravitational wave. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. A body reads the holder of the sign at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

5.7 The price of locality

Two consequences of the model's locality, stated against nature's measurements. Born's rule is the NodeReader's declaration, and exclusivity is derived from the count conserved under the credit's one draw, the NodeReader's declaration (Section 5.3). A clicked record's empty wave still writes its form into the holders it sources until the front reaches it: a Node with six Ports erases nothing beyond itself, and the NodeReader's front erases inside the click's light cone, never faster (Section 5.4, the open list). And a pair is one laid record, its phase a shared classical variable inside Bell's hidden variable [24], so every local credit of it gives $|S| \leq 2$ (THEOREM; CLICKS) [25]; the meeting through the root gives $S = 2\sqrt{2}$ with nothing signalled, resting on the credit rule, a declaration (ASSUMPTION, FITTED; Section 9.4). A which-way measurement without absorption is no third price and is not computed here. The model's opaque NodeReader in one gap loses the fringes and reads one gap's envelope, as nature's absorbing which-way markers do [1, what is open, item 4; the rows against nature, (g) and (h)][3, the click is the meeting of the future with the past; go over the paper and see which mismatches with nature the clicks resolve].

5.8 The formulas of clicks

Table 2 collects the formulas of clicks that the model gives, each with its status and the section that derives it [3, the click's algebraic definition; the mathematics of the bound bodies and how it bears on the clicks].

6 The families and the universe

The Universe24 it computes is declared family by family. A family is a band of the one rule, not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair $[\text{num}, \text{den}]$, with $\cos \omega_0 = \text{num}/\text{den}$, and its shape. The shape of a family of quanta is its dimension; the shape of a holder is its sources with its divisor. When a pair is written over the GameBoard's one denominator Γ , a mass is one integer m , with $\cos \omega_0 = m/\Gamma$ and the wall $6\Gamma^3$. Reducing a pair to lowest terms is a reading and never an act of the loader, so such families share one resolution of their remainders [1, the families from the rule; the mass is one integer; the wall is one].

How a family enters. A family enters the file as one row of the universe file, with two keys and nothing else: its pair and its shape. For a family of quanta the shape is its dimension, 1 or 2. For a holder the shape is its sources, which are the form, the form with the tensions, or the Wronskian, with the holder's divisor E_s and, for the massless holder, its rest. Its rotation at rest, its band, its coefficients and its wall, $w = 6 \text{den} \Gamma^2$, follow from Eqs. (1) and (4). Its parts number its sources, one real line per source. Its reach is set by its gap (Section 6.4). Its reads are those its declaration names, and gravity's and the binding holder's at the weight 1 (Section 3.4); it sources what it reads, at the weight it reads with. Every record clicks, a holder's included, since no row declares whether it clicks. Table 3 gives the rows of the universe, and Section 6.6 the numbers of the file [1, a family's declaration; the families from the rule][2, how to add a family].

Table 2 The formulas of clicks, each with its status. Every row is a formula of what a NodeReader reports, CLICKS. U is the clock's share and U_K the Newtonian potential the orbits measure (Section 8).

The formula	Status	What the model gives	Section
Born's rule, the count at a region	declaration; N_R and the scatter derived under it	$N_R = Q_R/W_c$, the share of the inflow through the region's front over the window; the pattern is in the histogram,	5.2, 5.3
The dead time	declaration	$Np_i \pm \sqrt{Np_i(1-p_i)}$ not derivable: the shares credit a region any whole number of quanta in one interval, and W_c over the inflow's rate is the mean spacing of the clicks and not a least one	5.3
The two-slit row	derived; computed	the near-field minima at the rows 15 and 33; the screen's inflow peaks at the interval 68.0 against the fourfold refined lattice's 61.8	9.2
The arrival and its spread	derived; computed	$t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$ Links per interval; the click's duration $\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band's curvature, 4.13 intervals at $k = \pi/4$ over $L = 34$ Links from the packet's centre to the screen; the centre's arrival 62.2	9.2
Bell's pairs	the credit declared (Born's rule for a pair); E and S derived under it	$E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$ at the ideal angles, 478/169 at the integer settings for equal parts	9.4
GHZ, three quanta	derived under the declared credit (S.40)	$E_3 = \cos 2(a + b + c)$, every pairwise E 0, Mermin's $M = -4$ where every local credit gives $ M \leq 2$	9.4
The clock's click rate	derived	$1 - f(\omega_0) \bar{U}$, $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, $1 - U$ at a small gap; the beat of two modes scales by N exactly at long wavelength, to a correction $(1 - N^2)(\omega_1^2 + \omega_1 \omega_2 + \omega_2^2)/24$ at finite rotations (S.31); a cavity of bodies the same, declared faces N^2	7.4, 8
The bending's clicks	derived under the one assumption; in U_K	$4U_b$; $(1 + \text{den}/\text{num}) 2GM_K/(c^2 b)$ against Kepler's potential	8
The absorption line	derived (S.16, S.43); the one-mode line under the pace read alone	parametric at $2\omega_b$, the growth $\epsilon/(4 \sin \omega_b)$; the transfer line at $\omega_j - \omega_i$, Rabi's rate, under both reads	7.4
Emission	derived	a steady rotation writes no light, nor a uniform motion in the long-wavelength band (S.42 (b)); a breathing body writes at its beat, at the first order in the breathing's depth	7.4
The which-way sum	derived	the continuity of the one inflow through the two fronts: a region behind one gap reads that gap's share and nothing of the fringe	5.7
The uncertainty relation	declaration, with its inequalities	$\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/N$ from the interval, $\sqrt{Np(1-p)}$ from the draw	5.3
No signalling	derived	$P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$, independent of b for parts of any amplitudes	9.4
A body's count in clicks	derived; the unit of credit declared	the share summed over the body's Nodes, what a NodeReader over all its Nodes credits; under a moving well it drifts by Eq. (13); a body's click one transition, $\hbar\omega$ per click	7.4
Magnetism	derived under the holder's rotation (S.33); not under the plain read	F_C/γ : the sign's halved current over the same wall as the time level, $E_s T$; 0.995 at 0.1c and 0.866 at 0.5c	8

6.1 The exact bands, and the third band

The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers only where $2 \cos \omega_0$ is an integer. So the rule carries exactly three exact massive rotations: $2 \cos \omega_0 = 1, 0$ and -1 , with the periods 6, 4 and 3 and the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM, GAMEBOARD). A record on an exact band, at rest in the vacuum, leaves no remainder in its rest rotation; so it never clicks and only turns. The exact bands are exact only in the vacuum. At a Node of pace p_0 the band's rest rotation is $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2$ with $g = (\text{den} - \text{num})/\text{den}$, which is an integer at $p_0 = \Gamma$ alone. So inside a well a record on an exact band leaves a remainder, and it clicks like every family's record [1, the exact bands; the exact bands are exact only in the vacuum].

The rule allows the bands $[0, 1]$ and $[-1, 2]$, and the universe does not use them [1, the third and the colours, closed as absent].

6.2 The rule's own universe

The universe of the model is the one in which every number is the rule's own. The pairs come from the integers 1, 2 and 3 of the three exact massive rotations and the three axes; every divisor is 1 except gravity's; and the remaining free numbers are named. Its rows stand in Table 3 [1, the families from the rule; the rule's own universe][2, the input].

The universe's integers are these. The Node clock is $\Gamma = 6,000$, a multiple of 3 so that the matter pair's $2/3$ is exact over it. The quantum's norm is $T = 2^{15}$, the normalisation of the quadratic form. Two bounds are not physics: the integers' width, 63 bits, and the Link unit, 16, the tension's resolution that sets the amplitude bound. Γ and T are units, as c and $\hbar$ are in nature. No line of Rule3 fixes T , since the rule is linear and the scale of its form is free. The energy line of Section 10.1 ties T to Γ , E_s and k_w only where a plane family reads the holder of the sign, and no family of this universe does. Every holder of the file is a record stepped by Rule3 at its pair and written by the one write. A file's starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final unification][3, two units and the bounds; nothing that is not Rule3 remains].

6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and by what a record can write, not by a name (DERIVED, GAMEBOARD). Rule3 reads two things of a Node and no third: the clock's pace p_0 , which is one number, and the Link's paces, which are the three directions of the rule's vector. So a holder read into the paces carries at most one time part and three axis parts, $1 + 3$. A record of one real line, neutral matter or light, has a form D and a stress and no Wronskian. A record of a plane,

Table 3 The rule’s own universe, its four families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape): each family with its pair [num, den] (in brackets over $\Gamma = 6,000$), what it holds at its divisor, its shape, what it reads, and its free number. A further family, the pair family of light, two real lines never summed, is declared by Bell’s file and not by this file (Section 9).

Family	Pair	Holds, at the divisor	Shape	Reads	Its free number
gravity, the time part and the three tensions	[1, 1] ([6000, 6000])	the content, E_g (1,000 in the bodies’ files, 10 in the gravity file; 10^6 in the three light-only files, gravity off in all but name)	1 + 3 real lines, sourced by the form and the three tensions	the content, its own level among it (the law’s line, implemented)	E_g , the hierarchy; its rest c_{vac} , 0 by declaration
the charge, the holder of the sign, whose quanta are light, born by the write	[1, 1]	the sign, 1	1 real line, sourced by the Wronskian of its readers	the holders of the content, at the weight 1; never its own level	k , the write weight, no key of the file (the plain write, $k = 1$)
the binding holder, not the strong force (Section 10.3); its own events have the gap	[2400, 2401]	the content, 1	1 real line, sourced by the form	the content, its own level among it (the law’s line, implemented)	the pair, its range $R = 20$ Links, a file’s choice
$\cos \omega = \text{num}_b / \text{den}_b$ matter, the scalar; a quantum that leaves a body is a message of matter	[m , den] ([4000, 6000])	nothing	dimension 1, one real line; a charged body of the same pair is a second row of dimension 2, a plane	the holders of the content, $x = (L_b + L_g) / \Gamma$; the plane also the holder of the sign	m , the mass

charged matter, has a Wronskian W as well. A holder is always real, one line per source, since a plane’s own W would be its own source. The shapes follow with no choice: *the scalar*, one real line, for matter, light and the binding holder; *the plane* for charged matter; *one plus three*, the time part with the three tensions, for gravity, which is sourced by the form and by the diagonal of the record’s own stress (Section 4.3); and *two real lines never summed at a Node* for the pair family of light in Bell’s file. The law as it stands takes no vector part of a holder and no off-diagonal stress, since Rule3 reads them nowhere. For a real record no read of a current exists within the three tests, so there is no frame dragging of neutral bodies (Section 10.3) [1, the final unification; the tension][3, every family has a dimension].

How gravity enters: a slowed clock, and no curved space. The model has no metric, no force and no law of motion. Gravity enters in three steps. (i) *The write*: at every Node, in each interval, every family of quanta writes its form over the action, $D_i \text{ div } T$, into gravity’s holder over the divisor E_g , by the second line of Eq. (1). (ii) *The holder*: gravity’s holder, [1, 1], steps by Rule3 at its pair [1, 1] and the divisor E_g . Around a static source it therefore rests at Poisson’s solution, $3G(r)s$, which is $3s/(4\pi r)$ far away. A change travels at one Link per interval, retarded. (iii) *The read*: every family reads the holder’s level ℓ at its Node into its own coefficients, Eq. (3), the clock’s pace once and each Link’s pace twice. What general relativity calls curved space stands here as the Link’s pace, light’s index $n = e^{2x}$; no space is bent in the model. The factor two on the Link is the one assumption of Section 3.4; the composition of the clock is the rule’s own [1, the paces; the clock once and the Link twice][1, the same clock and the same ruler for every act; the paces compose].

6.4 The reach of a holder, and the binding holder

A row’s rest solves $\text{num } \Delta^2 a - 6(\text{den} - \text{num})a = -3\text{den} \sigma$. This is its line with the source once an interval; Δ^2 is the six-neighbour second difference. Outside a body the rest therefore falls by $e^{-\kappa}$ per Link, with $\cosh \kappa = 3\text{den}/\text{num} - 2$, which is the family’s own band at zero rotation. So the gap is one number that is both the mass of the holder’s quanta and the inverse reach of the row (DERIVED, GAMEBOARD). [1, 1] has no gap, and its rest is Poisson’s over the whole GameBoard. [1, 2] has $\cosh \kappa = 4$ and falls by 0.127 per Link: a contact well with a reach of $1/\kappa = 0.48$ Links. A pair with num just under den has $\kappa^2 = 6\text{den}/\text{num} - 6$ to second order, and so a range $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links, which is Yukawa’s form [47] (S.11) [1, the well].

Assumption 4 (the binding holder and gravity’s divisor, decided with a reservation). The well of a record, $D_i \text{ div } T$, is laid into two holders. One is the binding holder: a family with a gapped pair [num_b , den_b], $\text{num}_b < \text{den}_b$, at the divisor 1. It is stepped by Rule3 at its pair, as the massless holder is, and started at its rest. With the gap, that rest is a sum with a range, $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$ Links, and $e^{-r/R}/r$ beyond it. It binds the bodies and is not the strong force (Section 10.3). Its own travelling events have the gap $\cos \omega = \text{num}_b / \text{den}_b$: a quantum with a mass whose Compton length is R , Yukawa’s own line [47] and no input. The other is the massless holder [1, 1] at the divisor E_g declared by the file: gravity, long-ranged and weak by E_g . Every family reads the holders its declaration names, at the weights it names, and sources each at the weight it reads with: gravity’s and the binding holder’s levels are read by every family at the weight 1. The two holders are among the readers, each reading its own level among the content. The nuclear holders of Section 10.3 are read by the nucleon families alone (the law’s line, Section 3.4; the selective read is a declaration, ASSUMPTION).

The first pair is [2400, 2401], with $R = 20$ Links, and the holder’s quantum has the rest frequency 0.029 per interval. The range is a file’s choice. An earlier version tied it to a cloud read as the proton, a reading the families replace (Section 10.3). In the implementation’s files E_g is declared large, 1,000 in the bodies’ files and 10 in the gravity file, so that gravity does not enter a body’s binding. Nature’s E_g is read from G , as the sign holder’s k is read from α . The hierarchy of gravity and electricity is the product of the two declared weights and the quantum’s energy against its unit of charge (Section 10.1). The decision carries a reservation: it may be wrong [3, the two rows][1, what is open, item 20].

6.5 The vacuum content

Assumption 5 (the vacuum content is 0, a declaration fixed by nature's measurements). The massless holder rests at a declared level everywhere: the vacuum content, the key **rest** of the universe file, which no line of the law derives. It is declared 0 where the light-only files carry the key, and absent, hence 0, in the universe file at $\Gamma = 6,000$.

A uniform rest c_{vac} is a change of units (Section 3.4), and only nature's measurements can tell it. Under the law's line every holder reads the content, as every family does. So the massless holder's own travelling events have light's paces identically in every level: gravity's events and light through the same wells arrive together, the difference of their two Shapiro delays is 0, and $\gamma_{GW} = \gamma_{EM}$. GW170817 bounds $\gamma_{GW} - \gamma_{EM}$ between -2.6×10^{-7} and 1.2×10^{-6} [48] (DERIVED from the law's line; CLICKS; S.12). Inside matter the source is read through the pace: gravity gravitates (DERIVED, GAMEBOARD). Since the two speeds are one speed, the measurement bounds no rest, and the vacuum content stays a declaration, 0 in the file (ASSUMPTION). The vacuum's write, which would give the faces a positive content, is a hypothesis of the law's documents (S.29) and is not claimed here. [1, every row reads the content; what is open, item 22][3, the vacuum content is 0, a declaration fixed by nature's clicks].

6.6 The file's numbers, and how the universe was chosen

The universe is chosen by one principle, stated before any row: one rule closes everything; the families are its inputs, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The road to them:

(i) a family is a band of the rule, and its row holds only its two keys; (ii) there are three exact massive rotations, the pairs [1, 2], [0, 1] and [-1, 2] (THEOREM, Section 6.1); (iii) the three of Rule3 is the three axes, so the massless holder of the content carries 1 + 3 parts and every other family is a scalar; (iv) light and matter have one denominator, their pairs written over Γ , a multiple of 3, and the binding holder's pair is the one pair not over it; (v) the pairs come from 1, 2 and 3 (ASSUMPTION): [1, 1] for gravity and the charge's holder, and [2, 3], the first pair past the exact bands, for matter, while nature's pair is far past it (Sections 8 and 10.1); (vi) every divisor is 1, under which the universe carried no free number but the mass; (vii) two numbers stand where nature's scale forced them, the binding holder's gapped pair and gravity's divisor E_g (ASSUMPTION, with the reservation of Section 6.4); (viii) the vacuum content is 0 where no write is declared (Section 6.5); (ix) the sign holder's write weight k is the plain write 1 and no key of the file, read from nature's α as E_g is from G (Section 10.1).

What remains is this: two units, Γ and T ; two bounds, the integers' width and the Link unit; and five numbers of physics, which are matter's numerator m , the binding holder's pair, gravity's weight E_g , the sign holder's write weight k and the vacuum content. Under the hypothesis of Section 7.5 there is one more, the rate of the vacuum write. Every other divisor is 1. The three light-only files read a copy of the file with the two holders of the content at the divisor 10^6 , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION). The electron's mass over Planck's and the proton's mass over the electron's are open and not derived (Section 10.3) [1, the rule's own universe; the final unification][3, the rule's own universe is the universe; the final unification].

7 Bodies

7.1 What a body is, and how it is computed

A body is its record over its Nodes, together with the quanta that record carries; it is never one Node. In plain words, a body is a record held at its Nodes by the holders' wells. Its count is the number of its quanta, and it clicks, as Section 5.4 says, when a quantum from outside meets its own transition. Its count is the share of its record (Section 4.4), a reading and not a state. Its form $D_i \text{ div } T$ is written into the holders it sources, and the holders enter Rule3 through the paces alone. The record is bound by the wells its form sources, that is, by the binding holder's screened sum with the range R ; gravity's $1/r$ is weak by E_g and enters no body's binding. No law is written for a body. A body is the support of a standing mode of the one rule, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named). (a) The body is the fixed point of its binding holder, standing where the sources return themselves. (b) A holder's level is the static solution of the row's line. The start solves that line from the bodies' forms to its own stop, after which Rule3 steps the holder. (The implementation's stop misses the line by up to one fine unit over $1 - \rho$ in the board's lowest mode, where ρ is the row's Jacobi factor; that is 137 fine units on the open 25-cube, a finding of the engine's register and not of the law.) (c) A body's quanta are its record's share, carried wherever its record's current carries it. (d) A free packet or a cloud falls as $a = \Delta |d\omega_b/d\ell| \nabla \ell$, where ℓ is the content level [1, what a body is; the four lines of the body][3, what a body is].

How a body is computed. The source of the held rows is the record's form $D_i \text{ div } T$, the law's line. The implementation's start laid the count, and the generator's lay under a level is being checked, with the write's factor N^4 read in the level and no physical return (the open list; GAMEBOARD). The body is computed by the rule's own iteration and by no other formula. Start from a spread of the declared count over the Nodes. Write the wells that spread lays into the holders, and let them rest. Find the record's top mode in those paces by Rule3's own read act, iterated. Scale its two levels so that the share carries the declared count. Write the wells that record lays. Repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the cloud and a narrow one the compact body, two fixed points at one count (the generator's readings of the earlier versions; no standing body in this version of the implementation). [1, the generator; the four lines of the body][3, the generator is Rule3].

7.2 Stability

Let a body of M quanta have the record ϕ over its region, with $\sum \phi_i^2 = 1$ and the counts $c_i = M\phi_i^2$. Let every holder r it sources be a hollow ($\sigma_r = +1$) or a hill ($\sigma_r = -1$), with the static kernel $\mathbf{G}_r$ of its line and the divisor E_r . The well at a

Node is then $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$. To first order in Φ/Γ the well raises the one-Node rotation $2 \cos \omega$ by $4g\Phi/\Gamma$, where g is the gap of Section 6.1. The body's fixed point is then a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{ den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and $dF/d\phi = 2\lambda\phi$ on the sphere is the fixed point's own equation, with $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$ (THEOREM, GAMEBOARD; S.13). Let $\phi_R(x) = R^{-d/2} \phi(x/R)$ be the body dilated, with d the GameBoard's dimension. The pressure is then K/R^2 , and each holder's well energy is W_r/R^{s_r} , where s_r is the holder's scaling power at the body's size: $s_r = d$ beyond the holder's reach, and $s_r = d - 2$ within it or for the massless holder [1, the functional and the dilation].

Theorem 4 (the stable body, under the dilation and to first order in the well). *At a fixed point of the binding row of width R , the virial $2K = \sum_r \sigma_r s_r W_r$ holds, and the second variation of F along the dilation is $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$. A body is stable only where this sum is positive. It is a minimum of F where, in addition, the Hessian on the modes orthogonal to the dilation and the translations is positive.*

(Proved in S.48, by Derrick's scaling [49] and Pohozaev's identity [50] on the GameBoard.) The theorem is a statement about the functional of Eq. (12), to first order in the well, along the one direction of the dilation, with the dilation taken as continuous. It is not a theorem of the integer step on the full lattice, and the Hessian beyond the dilation is open (Section 10.3).

The factor $s(2 - s)$ is +1 at $s = 1$, 0 at $s = 2$ and -3 at $s = 3$. So a hollow holds the body where its well falls more slowly than the pressure, and breaks it where the well falls faster. Three consequences follow. (a) One hollow beyond its reach binds every mass on a chain. In a box it has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless holder holds the body. Poisson's well in a box has $s = 1$, so $F = a/R^2 - c/R$ has one critical point, $R = 2a/c$, a minimum for every mass; this is Pekar's body [51]. Beside it, a hollow beyond its reach adds a barrier: $F = a/R^2 - b/R^3 - c/R$ on a Gaussian body, where the wide critical point is a minimum and the narrow one a saddle, and there is none where $3bc > a^2$, which is the collapse (DERIVED, GAMEBOARD; S.13). (c) A collapse ends in a frozen clock. Beyond first order the well saturates and the Link's pace weakens the pressure. Since no pace reaches 0 before the rounding, a collapse ends in a cluster of Nodes whose clocks are frozen to the unit, and never in a closed Link [1, the functional and the dilation][1, what is open, item 29].

7.3 The two branches, and the status of a standing body

The binding row's fixed point has two branches, a wide cloud and a compact pixel, and the start's width alone chooses between them (Section 7.2; DERIVED for the functional, S.13). No standing body is computed. The implementation's look at a laid body of 10,000 quanta on the open 25-cube, over 4,000 intervals, finds the generator's bodies bound but not standing within rounding (a finding of the implementation; GAMEBOARD). That reading, the three problems of the computation it names, none of them in Rule3 (the lay off the integer fixed point, the turn's convention and the rounding walk), and the fix proposed stand in S.62. The computed body is coarse in the lattice's pixel by construction (a few Links, about 10^3 levels per Node). So its drift and breathing are the lattice's budget and the lay's, and not the law's: $(\text{Link}/R)^2$ and $\sqrt{n}/A$, stated for each file. The first is 2×10^{-23} at nature's pair; the second is the engine's alone; and the law's derivations are of the line without the division. Until then the body's rows carry only what is computed: its definition (Section 7.1), the static fixed point and the modes by hand (S.46, S.60), the stability conditions as derived (Section 7.2) and its clicks (Section 7.4), and no sentence that a body stands under Rule3. The clouds of the earlier versions and their numbers belong to the law's document and not to this paper. The fix of 2026-10-03 refines the rest to the line within one fine unit at every Node by the scaled residual, 0.51 level on the 25-cube. It also makes the write's factor one rounding half up of the booking over the one wall, exact beyond the width. And it names the room's bound under the Link's factor, the room's factor of the ceiling $q_{\max}/P$, which its name owed before any file with tensions of either sign is laid. Every shipped file's number is unmoved, and the nine gates say MATCH. The standing body's run keeps its 48 images to the bit through 1,000 intervals; read 2's third part and read 3's growing ringing are named findings (GAMEBOARD) [2, the interval; the gates].

The atom: known from clicks, laid on the GameBoard. The atom is known from clicks: its spectra, its lifetimes and its scattering. Nothing of it is seen on the GameBoard. What is tested is whether the way from the clicks to the GameBoard and back carries the atom. This is a supporting finding, with its state named (S.46, S.52). An atom is a bound body of the holder of the sign. A nucleus, which is one quantum of its own family, writes the time angle $\theta_0 = Zac/r$, with $c = 1/\sqrt{3}$, into the sign holder's row by its Wronskian. Electrons, which are quanta of the matter family, read that angle into their rotation. The standing records that Rule3 makes in that angle are the atom's levels, Schrödinger's in the slow limit. With the nucleus's angle alone, the levels have Bohr's form, $E_n \propto 1/n^2$ with $a_0 = \sqrt{3}/(m^* Z \alpha)$. A record that read its own sign row (S.46 (c), the state before line (1) of the decision of 2026-10-02) had the law's form at large Z only, $1 - 1.25/Z$, and not hydrogen. That is the finding which line (1) answered: under line (1) a record reads every sign row but its own, and hydrogen's lines are those of S.60. The holders of the content keep the self-read (line (2)). The atom of this paragraph is derived and not yet run. No atom file is computed; the atom is closed by decision, with its derivations standing [3, the law's words]. What the implementation's design of an atom file found, that there is no admissible integer file at $\alpha_{\text{law}} = 1/28$ under its four constraints together, is recorded in S.62 (a named finding; GAMEBOARD). What a click at the atom gives is derived: the born light is at the resonance $\omega_e - \omega_g$ and not at the energy-conserving frequency, with the share's difference booked; absorption erases the photon's empty wave inside the click's light cone; and there is no recoil, consistently, since the write conserves no momentum and the frozen proton has infinite inertia, so nature's recoil shift is outside the claim.

Bohr's levels, the threshold, the lines and the two-mode binding are unchanged by it, and the cause of a click is outside the paper's claims. The lines are beats of two levels, and the mass defect is the click layer's reading of a frequency. The orbital Zeeman splitting has $g = 1$ (supporting findings with their numbers in S.46; CLICKS for the levels' ratios, GAMEBOARD for the records). The electron and the nucleons are quanta of their families. An electron is one quantum of count $\sin \omega_e$, ten orders below the compact threshold, and nothing smaller than a quantum clicks. Mass is the family's gap, $T \sin \omega_s$, bound or free. What decays is a body, from which a click at its front parts a quantum; a quantum itself never breaks, since it has no part that the click credits (DERIVED; CLICKS). The criterion is one line, a rule of reading: what nature splits with energy is read as a body on the GameBoard, and what no energy splits is read as a quantum of a family. The nucleus, the atom and the molecule are bodies by the one mechanism the law has for a body, a record standing in the well of a holder: the sign holder for the atom and the nuclear holder for the nucleus (ASSUMPTION) [1, the bound body is one Node; what nature splits is a body; the nucleus][3, the electron and the nucleons are quanta, not pixels; no mass without a bound body; a bound body that breaks up is an atom that breaks up]. A nuclear holder that the electron reads is excluded by the spectrum, by hydrogen's $1S$ and by muonic hydrogen (DERIVED for the shift's form, the exclusion a comparison; one derivation; CLICKS; S.46). So the nuclear holders of Section 10.3 are read by the nucleon families and not by the electron, a selective read that is declared. The table of S.60 sets the atom's clicks, the lines and the thresholds, against nature's under the two lines decided for the law, with how each was derived; this is a supporting finding and no more. Nothing of Bohr's numbers is claimed from a run (S.62). What the atom of the model does not give is listed with its roots in Section 10.3 and S.52 [1, the atom is a bound body of the holder of the sign].

7.4 A body's clicks

What a NodeReader reads of a body follows from the body's record at its Nodes, from the readings of Section 4.3 and from the paces, with no further line. Every formula below is a derivation, with its status and its fence; none is the result of a run [3, the mathematics of the bound bodies and how it bears on the clicks].

A steady rotation writes no light. Take a body at rest with the plane record $\text{re} = A \cos \omega_b t$ and $\text{im} = eA \sin \omega_b t$, where e is its ellipticity. Its Wronskian is $W = -eA^2 \sin \omega_b$ at every interval. So the source of the sign holder is constant, and its holder's level stands at its rest: no travelling events, no click. Rule3 keeps a body's total Wronskian, and keeps an isolated Node's exactly. So a one-Node source cannot breathe, and a body's breathing moves W between its Nodes. Its light is written by the second moment of its charge; the monopole is constant by the continuity identity. The light born therefore carries the frequency of the breathing and not the rotation, and the click rate of a far NodeReader is the share of that current through its front (S.42). So a symmetric body at rest clicks nowhere. A body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source GAMEBOARD, the far NodeReader's rate CLICKS) [1, light is born by the write].

The invariant, and the count's drift. For one Node of a body, $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with ω changing slowly keeps $A^2 \sin \omega_b$ invariant (S.14). This is the discrete adiabatic invariant of Rule3's line. The form $D = A^2 \sin^2 \omega_b$ and the share are not invariant; the recurrence stepped over 20,000 intervals keeps $D/\sin \omega$ to 4×10^{-5} . The share in quanta at a Node is $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b/T$ (Section 4.4). Under a moving well it therefore moves as

$$\frac{N(t_2)}{N(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{p_i^2(t_1)}{p_i^2(t_2)}, \quad (13)$$

with no current through any Link. The first factor is the rotation's own change under the clock's pace; the second is the Node's weight in the share (DERIVED for the invariant, to the first order in the slow variation, and for the drift; GAMEBOARD; S.14). In the model the invariant stands where nature's number of quanta, E/ω , stands, and the share stands where nature's energy stands. The same identity is Planck's relation on the lattice, $\text{share} = I \sin \omega$ with $I = A^2 \sin \omega/T$. So $E = \hbar \omega$ holds, with T for $\hbar$ and $\sin \omega$ in place of ω (DERIVED; CLICKS) [1, the count is the record's share; the paces' own change moves the share with no current].

The clock's click rate. A bound body's rotation reads the content once. At rest $\cos \omega_0(U) = \cos \omega_0 + 2gU$ to first order, so its rate falls by $f(\omega_0)U$, with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$: 1.063 at [2, 3] and 1 as the gap closes. In series, $f = 1 + \omega_0^2/12 + \dots$; this is the finite-rotation correction in ω of a rate that is exactly N in $2 \sin(\omega/2)$ (S.31). A light clock between two faces falls by $2U$, the Link's pace read twice (Section 10.2; S.15) (DERIVED, CLICKS).

The resonance. A bound record at the rotation ω_b that reads a light level into its content is modulated by $\epsilon = 4g(1 - 2x_0)b_L/\Gamma$. The resonant part at $\omega_L = 2\omega_b$ grows the amplitude at $\mu = \epsilon/(4 \sin \omega_b)$ per interval, inside the full width $\epsilon/\sin \omega_b$. A body with two bound modes $\omega_i < \omega_j$ transfers between them at the difference $\omega_j - \omega_i$, at Rabi's rate $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$ (DERIVED, two derivations; S.16). Under light's declared read, the holder's rotation (Section 5.5), the Link phases couple the modes through the inter-mode current. The coupling is resonant at the difference line for drives below $\omega_j + \omega_m$, with Rabi's rate $\Omega_R = \theta_L \omega_L |d|/2$ for a small line; it oscillates at every detuning with the envelope $\Omega_R^2/(\Omega_R^2 + \Delta^2/4)$. At the sum $\omega_j + \omega_m$ the same coupling drives the two senses together. That is a pair line, outside the absorption and emission of this section (the atom's lines at optical frequencies lie far below it). The one-mode line at $2\omega_b$ has no analogue there (DERIVED; S.43). The read's side is computed; the fraction absorbed, with the body's write back, is open (GAMEBOARD for the record, CLICKS for the body's share) [1, the rows against nature, (y)][1, the two-mode line].

A body's count in clicks. A body's count is its share summed over its Nodes, which is what a NodeReader over all its Nodes credits. Under a moving well it drifts by Eq. (13) (DERIVED, CLICKS). The implementation's declared region credits the share: a bolometer's reading, Millikan's slope 0. What clicks in nature is a body, and the model declares the click as one quantum's change in the NodeReader's own record. This is the NodeReader's fourth declaration, beside its window, its region and its floor (ASSUMPTION) [3, the whole of a click is the detector's own transition]. The NodeReader's

record answers at its rotation by the two-mode line above (S.43), one transition per click. The transition takes the share $T(\sin \omega_j - \sin \omega_i)$, which is one photon's share $T \sin \omega_L$ to the order $\omega_i \omega_j$, below 10^{-18} at nature's gap. So a body counts photons and reads $\hbar \omega$ per click, Millikan's slope T (DERIVED; S.14). The share stays the conserved form and the draw's measure, so every ratio of the checks is unchanged (THEOREM). The implementation credits the share; the declared unit is not yet implemented (CLICKS) [1, the count is the record's share].

7.5 Clusters: how they enter, and what each thing represents

How a cluster enters. The model writes no line for a cluster. One assumption names the universe it describes: every body is bound, no body is free, and a record that is in no body is a message between two clicks (ASSUMPTION). A cluster is bodies standing in one another's wells. Between two bodies there is nothing but the click, the light of the sign holder and the two real rows [1, the bound body is one Node; the universe is bound; the algebra of clusters].

The clicks between bodies are clicks in the one sense. A body is a region of Nodes with a front: the Ports where its tail meets its neighbour's. So a body is a NodeReader in the meeting's form. The click between two bodies is the pair (R, t) , one whole quantum credited to the region of the taker at the interval t , drawn by the share Q_R/W_c of the inflow through its front on the forward orbit. A body's count is its share summed over its Nodes. It moves only by the clicks at its front, up to the drift of Eq. (13) (DERIVED from the definitions; CLICKS) [3, the click's algebraic definition; the click is the meeting of the future with the past].

What each thing represents in a cluster. The three levels at a Node are the nesting. The binding holder's level is the matter's tail, the sign holder's level is the charge's well, and the massless holder's level is gravity's well. These are three kinds of binding by rank: quanta into a body, bodies into a charged cluster, charged clusters into a gravitational cluster (DERIVED, GAMEBOARD). The couplings are at the edge: every divisor is 1 but gravity's E_g , and the universe of clusters carries Γ , T and the two holders' weights and nothing else (ASSUMPTION for the divisors) [1, the levels at a Node are the nesting; three kinds of binding, not three levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the couplings are at the edge] [3, what a body is; the rule's own universe is the universe].

The clusters' cosmology. The universe of the model is a finite ball of bodies flying through the static lattice. Every body is pulled by the inverse square of the others at Kepler's G_K , Newton's constant as the orbits read it (Eq. (20); Section 8, (j)). A shell of dust at the radius a feels the mass inside it, so $\ddot{a} = -(4\pi G_K/3) \rho a$ and $\dot{a}^2/a^2 = (8\pi G_K/3) \rho - K/a^2$. These are Friedmann's dust equations, exactly, for the ball's interior [52], with $K > 0$ where the ball is bound. A receding source's redshift is $1 + z = 1 + \beta$ at first order: Doppler's and Hubble's known forms, derived here. The two derivations, the Doppler period of row (m) times the moving clock of row (e), agree exactly (DERIVED, two derivations; CLICKS; S.29). The paper's cosmology is claimed at $z \ll 1$ and not beyond. The second order, where the deceleration is read, is named among the findings of Section 10.3 [1, the clusters' cosmology].

8 The forms the rule arrives at, set beside nature's

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run. The form of nature beside it is never an input, except for the one assumption of Section 3.4. The coefficients contain no metric, no Lorentz transformation, no potential and no force. They contain only the paces of Section 3.4: the clock composes by the rule, and the Link is the clock twice by the one assumption. $U = c/\Gamma$ denotes the model's potential at the point named, the clock's share per level, and U_b its value at the impact parameter b [1, the rows against nature].

Two potentials. The forms are stated in two potentials. The model's own potential is the clock's share, $U = c/\Gamma$, the content x that the paces read; no NodeReader reads it. Every test of gravity takes GM from Kepler's law of the orbits of matter and c from light, so it reads $U_K = GM_K/(c^2 r)$. In the model a free packet of matter of the pair [num, den] falls in a content gradient at $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla U|$, exactly from its band, Eq. (4) (S.18). So the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes.} \quad (14)$$

Each form below is lettered as the law's rows are. It is written first in U , where the composed paces give, in the post-Newtonian words, $\gamma = 1$ exactly and $\beta = 1$ at the small gap, the metric $N^2 = e^{-2U}$ and $h^2 = e^{2U}$, and the redshift's factor $f(\omega_0) = 1 + \alpha_U$ with $\alpha_U = 0.063$ at $[2, 3]$. It is then written in U_K , where the same algebra gives $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$ and $1 + \alpha_K = \tan \omega_0 / \omega_0$: 1.5, 1.25 (the small-gap closed form under the quadratic band; the lattice band's advance is in (d)) and 1.329 at $[2, 3]$, and 1 as the gap closes. The two columns are one algebra, and nature's tests read the second (DERIVED, CLICKS; S.18, S.19) [1, the rows against nature] [3, general relativity in the law].

- (a) *The redshift* (DERIVED, CLICKS). The clock's pace is $p_0/\Gamma = (1 - 1/\Gamma)^\ell = e^{-U}$ exactly, with $U = -\ell \ln(1 - 1/\Gamma)$, which is ℓ/Γ to the first order in $1/\Gamma$. So the clocks of two Nodes tick in the ratio $e^{-(U_1 - U_2)}$, $1 - (U_1 - U_2)$ to first order. The rest rotation of a bound record, the clock this row names, shifts by $f(\omega_0)U$ (Section 7.4): $1.063U$ at $[2, 3]$ and U as the gap closes. Against the Newtonian potential that the orbits measure, it is $z = (1 + \alpha_K)U_K$, 1.329 at $[2, 3]$, where nature's clocks give $|\alpha| < 10^{-4}$ [38]. Beside it: Pound and Rebka's line [56] [38].
- (b) *The bending of light* (DERIVED, CLICKS). At a Node whose clock's share is $U = \ell/\Gamma$ (the content level over Γ , Section 3.4), light's index is $n = e^{2U}$, $1 + 2U$ to first order, because the Link's pace reads the content twice. Far from a body, its level is the massless holder's rest, $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$, where σ_{tot} is the body's wells per interval over the holder's divisor E_g . A beam passing at the closest distance b turns by the transverse gradient of the index, integrated along the path (S.20):

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi \Gamma b}, \quad (15)$$

twice the angle $2U_b$ that a Newtonian pull of the same potential gives. The factor two is the ruler going with the clock, $hN = 1$, the one assumption of Section 3.4. Against the Newtonian potential that the orbits measure, the bending is $(1 + \gamma_K) 2GM_K/(c^2 b)$ with $\gamma_K = \text{den}/\text{num}$, 1.5 at [2, 3], where Cassini's tracking gives $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [38][57]. Beside it: Einstein's $4GM/(c^2 b)$ [38]. The bending is the $\beta = 1$ case of the push on a moving record:

$$\mathbf{F} = \frac{E^2 + P^2}{E} \nabla U = E(1 + \beta^2) \nabla U, \quad \beta = \frac{P}{E} = \frac{v}{c_{\text{local}}} \quad (16)$$

with $E = \omega$, $P = cN^2 k$ the momentum at the composed paces, and β the record's speed over the local light speed cN^2 . At rest the push is $E \nabla U$; for light it is $2E \nabla U$, and Eq. (15) is its $\beta = 1$ case. Its bending is exact in the gap, while the orbit's second order at the computed pair carries the band's k^4 term (S.19) (DERIVED from Eqs. (3) and (4) under the one assumption, fitted; S.58) [1, the push on a moving record]. The paper's momentum is the wave number $\mathbf{k}$, so the lattice form is $(E^2 + c^2 \mathbf{k} \cdot \mathbf{k})/E$ with $c^2 = 1/3$. Written with the velocity $\mathbf{v}$ in Links per interval, $\mathbf{p} = E\mathbf{v}$, it is $E(1 + 3v^2)$ at $N = 1$, since $\beta^2 = v^2/c^2 = 3v^2$; the post-Newtonian factor $1 + \gamma\beta^2$ has $\gamma = 1$ here, derived. At the computed pair the coefficient of β^2 is $\gamma_K - \alpha_K/2 = 1.335$ and the rest push is $f(\omega_0) E$. As physics it is derived here, and it coincides with the known post-Newtonian deflection factor $1 + \gamma\beta^2$ at $\gamma = 1$, derived and not declared. Nature is met by its click formula, the ratio $1 + \beta^2$ of the deflection angle to Newton's (CLICKS).

- (c) *The delay of light* (DERIVED, CLICKS; S.20). Light's coordinate speed at the content U is $e^{-2U}/\sqrt{3}$, so its index is $1 + 2U$ to first order. Take a signal grazing a body at the impact parameter b from x_1 to x_2 , with $U = \mu/r$, $\mu = GM/c^2$ in Links and b , x_1 , x_2 in Links. It is late by $2\sqrt{3} \mu \ln(4x_1 x_2/b^2)$ intervals: Shapiro's delay $(1 + \gamma)(GM/c^3) \ln(4x_1 x_2/b^2)$ at $\gamma = 1$. Beside it: Shapiro's delay [58].
- (d) *The perihelion* (DERIVED, CLICKS). A slow body's orbit in the composed clock's metric, $N^2 = e^{-2U}$ and $h^2 = e^{2U}$, is the post-Newtonian orbit with $\gamma = 1$ exactly and $\beta = 1$ in U for the quadratic band, the small-gap limit. So the periastron advances by $6\pi\mu/(a(1 - e^2))$ per orbit, with $U = \mu/r$, μ in Links, a the semi-major axis in Links and e the eccentricity. This is Einstein's value, which at nature's pair is Mercury's 43 arcseconds per century (a comparison). At the computed pair the band's k^4 term, of the same order as the relativistic one, adds of the order of ten percent to the advance; the addition depends on the orientation and vanishes as $\tan^2 \omega_0$ (DERIVED at the small gap; the quadratic band an ASSUMPTION at the computed pair; S.19, S.34). Against the Newtonian potential that the orbits measure, the advance is $(2 - \beta_K + 2\gamma_K)/3$ of Einstein's, with $\beta_K = (\text{den} + \text{num})/(2\text{num})$, 1.25 at [2, 3], the small-gap closed form under the quadratic band (with the lattice band the advance gains of the order of ten percent, orientation dependent, and no single β_K stands for it). In nature the planets' and the Moon's orbits give $|\beta - 1| < 10^{-4}$ [38]. Beside it: Einstein's $6\pi GM/(c^2 a(1 - e^2))$ [38].
- (e) *The moving clock: the band's kinetic scale and the fourth order* (DERIVED, CLICKS; S.21). The tick's factor of a packet moving at the velocity v is $f = \Omega(k)/\omega_0$, where $\Omega(k) = \omega(k) - k\omega'(k)$ is the rotation at the moving centre. From Rule3's exact band, $f = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0) v^4/(8c_m^4) + O(v^6)$, with $c_m^2 = \omega_0/(3 \tan \omega_0)$ the band's kinetic scale. The coefficient of the fourth order is 1.6926 at [2, 3] and 1 as the gap closes (Fig. 5). The kinetic scale $c_m^2/c^2 = \omega_0/\tan \omega_0$ ($1 - \omega_0^2/3$ at a small gap) is 0.752 at [2, 3], so c_m is 13 percent below light's. The band's largest group speed, along the cube's diagonal at $\mathbf{k} = (\pi/2, \pi/2, \pi/2)$, is $c \cos \omega_0$, $2/3$ of light's at [2, 3] (S.21); along an axis it is $1/4$ Link per interval, $\sqrt{3}/4$ of light's. At nature's matter pair c_m lies below light's by $\omega_e^2/6$, 2.2×10^{-20} , and the diagonal's largest group speed by $1 - \cos \omega_e = \omega_e^2/2$, 6.7×10^{-20} , at the LHAASO bound; at the axis bound these are 8×10^{-29} and 2.4×10^{-28} (Section 10.1). The electron is subluminal. Nature bounds the difference of matter's and light's limiting speeds with its sign: the subluminal case at about 10^{-20} by the Crab's synchrotron [61], the superluminal at 5×10^{-21} [62], and the earlier bounds at 10^{-15} to 10^{-20} [59]. The subluminal bound, where it holds, binds ω_e near 10^{-10} . Beside it: Lorentz's factor, reached as $c_m \rightarrow c$ [3, the click].
- (h) *De Broglie's relation* (DERIVED; GAMEBOARD for the band, CLICKS for the fringe spacing that reads it; S.22). A free record of a matter family has the band $\cos \omega_b = (\text{num}/(3 \text{den}))(2 + \cos k)$ and the group velocity $v = (\text{num}/(3 \text{den})) \sin k / \sin \omega_b$. To second order, $k = \omega_b v / c_m^2$, with c_m the band's kinetic scale of (e). Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS). In a pace gradient a free packet or a cloud accelerates toward the well at $a = \Delta |d\omega_0/d\ell| (\ell_+ - \ell_-)/D$ Links per interval squared, where Δ is the top velocity of its mode in Links per interval and $\ell_{\pm}$ is the content level at its two faces, D Links apart. So bodies with the same $\Delta d\omega_0/d\ell$ fall alike. Beside it: Galileo's fall [38]. The binding by the holders of the content gravitates as in nature: two bodies of one count and different binding fall alike. The seat of the electromagnetic binding energy is open (DERIVED; CLICKS; S.38). Between families: a free packet of the pair [num, den] falls at $c^2(1 - \tan^2(\omega_s/2)) \nabla U$ exactly, so two families of different gaps fall apart by $(\omega_n^2 - \omega_p^2)/4$ at small gaps. With the proton and the neutron as two families, beryllium against titanium gives $\eta = 32.3 \omega_e^2$, below 4.4×10^{-18} at the LHAASO bound. That is under nature's 10^{-13} for that pair [38] and 10^{-15} for titanium against platinum [37] (DERIVED, CLICKS; S.47) [1, the rows against nature, (i)].
- (j) *Newton's pull and orbits* (DERIVED, CLICKS). The massless holder rests at Poisson's solution with the sources, and at Laplace's outside a body: the content level is $\ell(r) = 3s/(4\pi r)$ levels at r Links, where s is the body's write per interval over the holder's divisor. So the acceleration falls as $1/r^2$, and a circular orbit's v^2 as $1/r$. Beside it: Newton's pull and Kepler's fall-off.
- (m) *A free record keeps its wave number* (THEOREM, GAMEBOARD). Rule3 is invariant under translation. So a free record keeps its wave number and rotation exactly, and a radiating body's period is read unchanged at every distance. Beside it: Hubble's law as Doppler clicks of matter flying apart through the static lattice, with no stretching of space (Section 7.5).
- (p) *The lens: a content bends light and matter alike, by one formula* (DERIVED, two derivations; GAMEBOARD for the index, CLICKS for the focus and the clicks; S.28). A stationary boundary conserves the rotation. So, for every family, a record

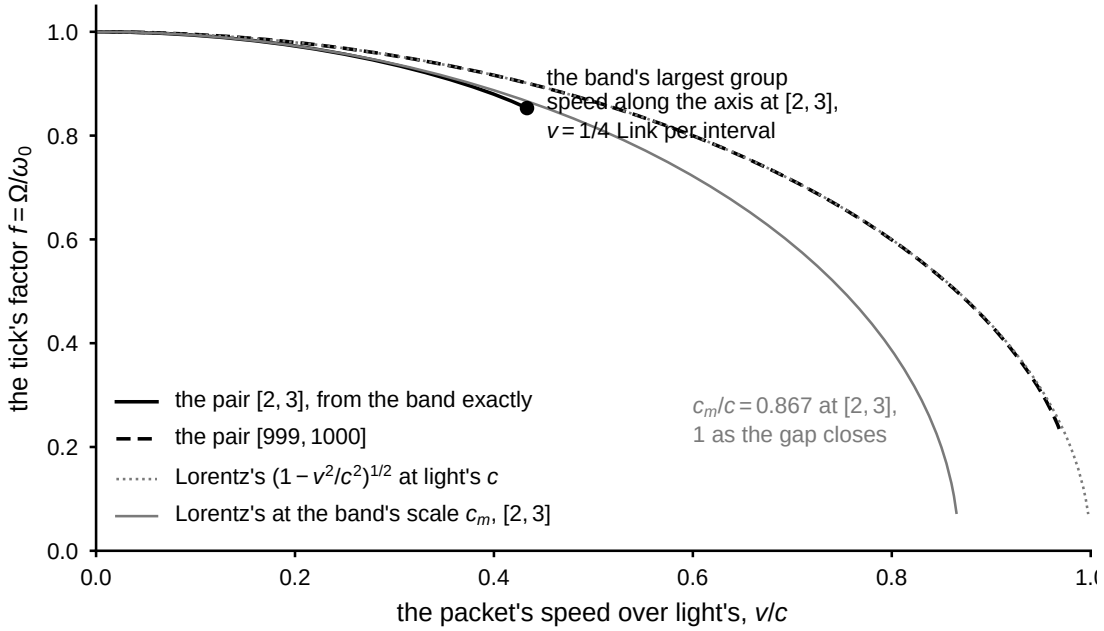


Fig. 5 The moving clock's factor from the band (Section 8, (e); DERIVED, CLICKS; drawn from the formula, no run). A packet of a matter family moving at the velocity $v = \omega'(k)$ along one axis ticks at the rotation at its moving centre, $\Omega(k) = \omega(k) - k\omega'(k)$, so its tick's factor is $f = \Omega/\omega_0$. It is drawn exactly from the band for the pair [2, 3] (solid), up to the band's largest group speed along the axis, 1/4 Link per interval (the dot), and for the pair [999, 1000] (dashed). Beside them are Lorentz's $\sqrt{1 - v^2/c^2}$ at light's $c = 1/\sqrt{3}$ (dotted) and at the band's own kinetic scale c_m , $c_m^2 = \omega_0/(3 \tan \omega_0)$, $c_m/c = 0.867$ at [2, 3] (grey). The computed pair's clock follows Lorentz's at c_m to the second order, and the small-gap pair's clock follows Lorentz's at c . The factor is a property of the clicks; no boost is a symmetry of the lattice.

of wave number k entering a content U takes the wave number k_{in} given by

$$1 - \cos k_{\text{in}} = \frac{1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}}{N^4}, \quad N = e^{-U} \quad (17)$$

light's index $k_{\text{in}}/k \rightarrow e^{2U}$ at long wavelength (nature's $n = 1 + 2\Phi/c^2$). It is exact in the law's own variable as $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$, with the lattice's chromatic term $k^2(e^{6U} - e^{2U})/24$ at finite k . Slow matter's index is $n_m^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$, growing as $1/k^2$ (nature's $n^2 = 1 + 2\Phi/v^2$, with the Kepler column's ratio at the computed pair, Eq. (14)). *What it does:* a slab of content bends every record toward its thick side, light a little and slow matter much. The fall of a body is the refraction of its own record. A window of content thick in the middle is a lens, and it gathers the record's share at a focus. *One quantum through it* (Section 5.4): the whole record passes the lens, and its share gathers at the focal spot. The one quantum clicks once, at one Node of the spot. The NodeReader draws that Node with its seed over the spot's shares and writes there: the levels that arrived are taken, and the count goes down by one. The next quantum draws again. The spot is the accumulation of the clicks, one per quantum, the image built click by click, as nature's single-photon images are (the formula derived from Eq. (4) at the composed paces; the one click per quantum the credit's declaration; the write at the one Node the NodeReader's declaration, its implementation pending) [1, the content's index for every family; the rows against nature, (p)][3, one quantum through a lens, a beam splitter and an interferometer is the law's own].

What goes in and what comes out. The model puts in one rule, the paces' read under the one assumption of Section 3.4, and the numbers of the run files. In the model's own potential the bending (b) and the delay (c) are Einstein's exactly, and the perihelion (d) is Einstein's at the small gap; $\gamma = \beta = 1$ in that column as the gap closes, light's ray metric. The post-Newtonian column at the computed pair departs with the gap. So at the rule's own matter pair the model does not describe the gravity of the solar system. The closing gap is nature's pair, where $\gamma_K - 1$ is below 10^{-19} at the LHAASO bound and 2.4×10^{-28} at the axis bound (Section 10.1); whether the model's bodies stand at that pair is not shown. Read as a metric, the composed clock gives $N^2 = e^{-2U}$ and the ruler $h^2 = e^{2U}$, against the isotropic Schwarzschild chart's $1 - 2U + 2U^2$ and $1 + 2U + 1.5U^2$: the spatial second order is 2 against 1.5. One click in the deep level reads the difference. Light's index e^{2U} places the photon sphere at $r = 2m$ with $U = m/r$ (S.30), so the capture radius of a black hole's shadow is $b = 2em = 5.44m$ against Schwarzschild's $3\sqrt{3}m = 5.20m$, 4.6 percent larger, an explicit prediction (DERIVED under the one assumption, which is fitted; CLICKS; S.30). Beside it: the Event Horizon Telescope's shadows. Sgr A*'s fractional deviation from Schwarzschild is -0.08 ± 0.09 [63], 1.4 standard deviations from the prediction under the VLTI prior, and M87*'s is -0.01 ± 0.17 , within 0.3 [64]. Rule3's step times the record's conjugate gives an exact continuity identity: the sign's density and its halved Link current are one conserved four-current, and the time turn with the three Link phases is one covector by minimal coupling. So the odd lines are sourced by the halved current over the same wall as the time level, with no number matched, and a co-moving reader feels Coulomb's force over γ (DERIVED; CLICKS; S.33). A source and a

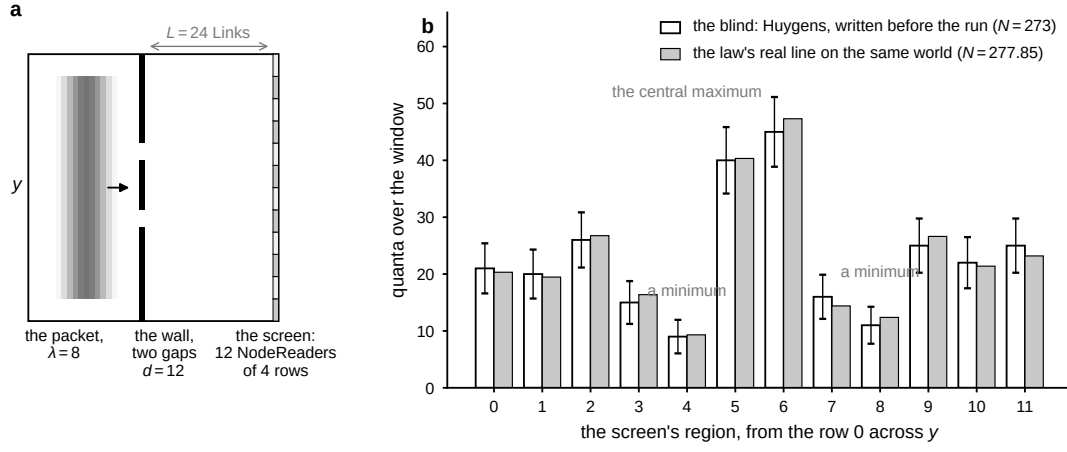


Fig. 6 The two slits, the one computed file (Sections 9.1 and 9.2; COMPUTED, CLICKS). (a) Universe24 as its file declares it. The flat GameBoard has 45×48 Nodes. The packet of light, at the wavelength 8 Links, is laid over the columns 4 to 16 (the grey raised cosine) and moves along $+x$. The wall at the column 20 has two gaps of three rows, 12 Links apart. The screen at the column 44, 24 Links from the wall, is twelve declared NodeReaders of four rows each. (b) The screen's row per region over the window of the intervals 45 to 130. The Huygens blind, written before the run, is hollow ($N = 273$, with the draw's scatter $\sqrt{Np(1-p)}$ per region). Beside it, in grey, is the law's real line stepped on the same file by the paper's script `two_slits_real_line.py` ($N = 277.85$ in the engine's labels, 278.0 in the physical ones). The run of the implementation gives $N = 278$ (Section 3.7). The numbers drawn are the blind file's and the script's; nothing is read from a run of the engine for the drawing.

reader of one family see one light speed, the band's own ($c_s^2 = \text{num}_s / (3 \text{den}_s)$) in the variable $2 \sin(\omega/2)$ and c_m^2 in ω , one as the gap closes; $1/\gamma_s$ uses c_s). In nature light's c is the same to 10^{-20} at the LHAASO bound [1, the odd line's source is the sign's own current; the rows against nature, (b2)].

9 The two slits and Bell: the one computed file and the check, derived and read by the implementation's checks

The paper contains no experiment against nature. The files below declare the Universe24 it computes; their results are derived from the rule and read by the implementation's checks (Section 3.7). The two slits is the one computed file set against a blind written before its run. Bell's file is a check of the declared credit and not a test of it: its number is fixed by the credit and by the determinism of the parts. Nothing in a file's lay, geometry or NodeReaders is built to bring the right result. The credit of Bell's reader is the declared and fitted exception, and Section 9.4 states it as such [3, the paper; one experiment and one gate; nothing in an experiment brings the right result by construction].

9.1 The two slits: the file

The source is a declared packet of light. Its wavelength is 8 Links along $+x$, $k = \pi/4$. Its levels are a raised-cosine packet of about ten quanta per Node, $b = 1,328$, over $x = 4$ to 16, flat across $y = 4$ to 44. The wall is the inner face at $x = 20$ with two gaps of three Nodes, $y = 17$ to 19 and 29 to 31, $d = 12$ Links apart. The GameBoard is flat, $45 \times 48 \times 1$ with the third axis folded, its y faces open and its x faces receding. The screen is the last column, $x = 44$, $L = 24$ Links from the wall. It is twelve declared NodeReaders of four rows each (Fig. 6a). The run is 130 intervals, read over the intervals 45 to 130. A record uniform along the folded axis has the three-dimensional band at $k_z = 0$, so the flat GameBoard carries the file's band exactly (DERIVED, GAMEBOARD). On this file the law's line gives 278.0 quanta through the screen in the physical labels (277.8 in the implementation's labels). The peak of the screen's inflow is at the interval 68.0, against 61.8 on the same file with the lattice refined fourfold (COMPUTED, CLICKS) [1, the rows against nature, (g)][3, the two slits; one experiment and one gate].

9.2 The two slits: what the model says it must give

The one file is read three times: by the prediction of the known forms, Huygens's sum and Young's spacing, applied to the geometry of the declared file and written before the run, and by the law's real line computed beside it. (a) *The fringe*. The spacing of the far field is $\lambda L/d = 16$ Links. At $L = 2d$ the fringe is the near field's, with the minima near the rows 15.3 and 32.7; by the sums of the regions these are the screen's regions 4 and 8 (DERIVED, CLICKS; S.23). Light's line at the vacuum's paces, stepped in real arithmetic on the declared file, gives 278.0 quanta through the screen (Section 9.1), region by region as the script's table lists (COMPUTED, CLICKS; the script of Section 3.7). The Huygens sum of the two gaps gives 273 with the same central fringe (Fig. 6b). The run's 278 is five quanta over it, 0.3 of $\sqrt{273} = 16.5$; the shares deviate from the blind by 0.041, and the extrema fall at the blind's regions. This is a comparison and not a statistical test, since the blind is Huygens's sum and not a draw. (b) *The clicks*. The screen's shares, rounded to whole quanta, are the expectation. The row drawn by the shares at the declared seed is the clicks, with the draw's scatter $\sqrt{Np(1-p)}$ per region. On the Huygens row at $N = 273$ the central maximum stands above its two minima by 5.7 times its scatter. (c) *The lattice's own number* is the inflow peak's 68.0 against 61.8 (Section 9.1), which the click line resolves (COMPUTED, CLICKS). [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

9.3 Bell: the file

The entanglement the model computes is one event with two trees. A pair is two records from one root. Each is read in two directions: forward, the future, and backward, the past. A click is where the two readings of one record meet. The pair's correlation is the partner's record read through the same root, re-weighted by what was realised. The pair is laid as the pair family: two real lines of light, never summed at a Node, with equal parts. Each side's NodeReader is a region at least half a wavelength across. It reports the signed level sums of the two parts over its region and knows nothing of angles; the ports enter at the credit. The reader applies the declared setting, a pair of integers (p, q) , with the port coefficients $e(+) = (p, q)$ and $e(-) = (-q, p)$. These are exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are $(1, 0)$, $(1, 1)$, $(12, 5)$ and $(5, 12)$, at 0, 45, 22.62 and 67.38 degrees. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \sum_{\text{window}} \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (18)$$

with α_k and β_k the amplitude of the part k at the port on each side, and $e_k(p)$ the port's integer coefficient on the part k . The window's sum stands outside the square, as the implementation's gate computes it, on both members of the level pair. The joint share is never the square of a local sum; such a sum one NodeReader could form alone, and it gives S at most 2. One draw is made per pair. It is the NodeReaders' draw with the declared seed, and the GameBoard draws nothing: the first side is drawn from its marginal and the partner from the conditional. Each side's write at the Node of its exchange spreads by Rule3 and changes no number of the row, one click per record (Section 5.4) (ASSUMPTION, CLICKS) [3, Bell in nature's own layout].

9.4 Bell: what the model says it must give

(1) Rule3 gives each part's amplitudes at the ports. The two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, and they are identical bit for bit (THEOREM, GAMEBOARD). (2) The meeting, with the linearity of Rule3, gives the joint amplitude. The realised click at A's port weights the parts by A's analyser. It is read back to the root, where the two parts were one event, and forward to B, where B's analyser weights them. The root sums the two branches: $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)$, with c the parts' laid weights, $(1, 1)$ for the symmetric lay (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [44]. Nothing in Rule3 forces it. Everything after it, the fractions, S and GHZ's M , is derived in the algebra (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase θ uniform over the pairs, A's + port at a and B's at b ,

$$\begin{aligned} \text{the meeting:} & P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} & P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} & E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the contrast credit, one side weighted:} & E = \cos 2(a - b) \text{ at one side's efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (19)$$

The second, third and fourth lines follow by averaging over the pair's phase θ (DERIVED, CLICKS; S.10). The fourth counts one side weighted by its contrast and always counts the other side's sign, by $\langle \cos x \operatorname{sgn} \cos(x - 2\delta) \rangle = (2/\pi) \cos 2\delta$; the symmetric readings give no cosine (S.10). So $E(a, b) = \cos 2(a - b)$ under the meeting. For the declared pairs it is exact, by Lagrange's identity of the reader on equal parts (S.10). At the four settings, then, $E = 119/169, -119/169, 120/169$ and $120/169$, and $S = 478/169 = 2.8284$ exactly. The efficiency is 1 by construction, and the marginal is one half on each side whatever the partner's setting (COMPUTED, CLICKS; S.10). For unequal parts, with r the ratio of their products at the ports, the same credit gives $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ with $\rho = 2r/(1 + r^2)$, and $S = \sqrt{2}(1 + \rho)$ at the ideal angles (DERIVED; CLICKS). The maximum of S over the four settings is $2\sqrt{1 + \rho^2}$: Tsirelson's bound [78] at $\rho = 1$, and never the algebraic 4. The declared pairs' 478/169 sits below $2\sqrt{2}$ because the angles are rational (DERIVED, CLICKS).

Without the declaration the GameBoard gives the second and third lines of Eq. (19), two results on local credits on the same file (DERIVED under the uniform phase, CLICKS). A credit of each side by the shares of its ports gives $S = \sqrt{2}$. A credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's bound. Bell's value therefore sits in the credit rule, a declaration, and it is no independent evidence that the rule accounts for entanglement. The same credit on four lines gives the three-quantum GHZ correlation [77], $E_3 = \cos 2(a + b + c)$, with every pairwise E equal to 0 and Mermin's $M = -4$, where every local credit gives at most 2 (DERIVED under the declared credit, CLICKS; S.40) [1, the GHZ gate]. No entanglement other than the pair's and the swap's is named as met (S.51). Between separately prepared bodies joined by a coupling or by a transferred quantum, the law's credits give at most 2. Nature's gate-joined ions [32] and transmons with a transferred photon [34] exceed it: a finding of the first rank against the law as written (S.56 (14)). No signal passes between the sides, since each side's marginal is one half. No settings are chosen by the GameBoard's own past, so a superdeterministic world [65] is not this one [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; Bell in nature's own layout].

10 Discussion: the constants, the calibration and the findings

10.1 The constants

In the Universe24 it computes, the quantum of action is one click, whose action is the quantum's norm T and whose energy is $T \sin \omega$ at the frequency ω , T at the band's top. Newton's constant is not declared. A body whose wells are s quanta per

interval makes the rest of the massless holder $3G(r)s$ at r Links, over the holder's divisor; far away this is $3s/(4\pi r)$. The level slows the clock by the potential $\Phi = -\ell/\Gamma = -U$, with Newton's sign, so (DERIVED, GAMEBOARD; S.25)

$$\boxed{\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi\Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi\Gamma E_g} \text{ per quantum of well, in Links and intervals}}, \quad (20)$$

a reading of Γ , the file's E_g and the lattice. Kepler's G , read from the orbits of matter as clicks, is $[2\cos\omega_0/(1+\cos\omega_0)]G_{\text{clock}}$ by Eq. (14), which is 0.8 of it at $[2, 3]$ (DERIVED, CLICKS). The vacuum content is 0 in the file, a declaration that no click bounds (Section 6.5). A cosmological constant, if the law has one, would be the rate of the vacuum write, a hypothesis not claimed here (S.29), and not this rest [1, the constants].

The fine-structure constant, a coefficient of the file; the units. Under the holder's rotation (Section 5.5) the quantum of a family is one unit of the invariant, $2A^2\sin\omega_s = T$. Its energy is $T\sin\omega_s$, and its Wronskian is $A^2\sin\omega_s = T/2$ for every family; this is the unit of charge. Charge universality is therefore derived from the writes: $|q_p| = |q_e|$ with nothing declared, which nature's measurements give to 10^{-21} (DERIVED; CLICKS; S.26). The holder of the sign carries one declared integer weight k in its write, as gravity's row carries E_g : the written level is k times the plain one, and the turn is the level over Γ . The law's coupling between two quanta is then $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$, independent of the family's gap (DERIVED under the rotation, k ASSUMPTION; CLICKS; S.26). The force on a body of n quanta and sense $\pm$ is $\pm n\nabla\theta$: Coulomb's q_1q_2 with $q = \pm ne$, where e is one quantum's Wronskian $T/2$. The reader sources the holder it reads, so the pair is reciprocal (DERIVED; S.26, S.43). Nature's $\alpha = 1/137.036$ fixes $k/(\Gamma E_s) = 1/28.33$ and nothing else. So α is a coefficient of the file, read from one click as G is; it is neither derived nor a finding. A charged body and its light conserve energy by a line of the law, $E_sT\cos\omega_s = k_w\Gamma$: the body's loss under the turn and the holder's gain under the write agree if and only if it holds (DERIVED for the identity under the implementation's present convention of the turn; the line ASSUMPTION; S.26). Under it $\alpha_{\text{law}} = (3\sqrt{3}/8\pi)T\cos\omega_s/\Gamma^2$, and at nature's gap $T = \Gamma^2/28.33$ (a calibration from α); E_s is then a unit of the level, and the file has one integer fewer. The line binds the plane families that read the holder of the sign, and not light or a real-line family. The rule's own universe has no plane family, so its $T = 2^{15}$ stands. The implementation's loader now refuses a body of a family that reads a holder of the sign at a count above 1, so that many quanta of charge are that many records of count 1; the charge rows' derivations stand under that line, per record and additive over records (one derivation; S.26). The four charge files of earlier versions, laid at 10,000 quanta per record, left the repository with the gate and their blinds are retired; a count-1 charge file is owed and not laid [1, no record reads its own write of the sign]. The two held rows are one kind of row with one kernel: the sign holder reads a quantum's Wronskian $T/2$, and the content holder reads its form $\sin\omega_s$. So gravity's coupling between two quanta of the family s is $\alpha_g(s) = (3\sqrt{3}/4\pi)\sin^2\omega_s/(\Gamma E_g)$, and the ratio of the two forces at that family is $F_e/F_g = kE_g/(2E_s\sin^2\omega_s)$. Between two families the weights cancel: $(F_e/F_g)_e/(F_e/F_g)_p = (m_p/m_e)^2$ with no number, and nature's $4.17 \times 10^{42}/1.24 \times 10^{36} = 3.37 \times 10^6 = 1836.15^2$ to the three digits of the inputs. This is an identity forced by charge universality and $F_g \propto m^2$, and no test (DERIVED; CLICKS; S.44). On the board there is no 10^{42} . The number appears when a NodeReader compares the two forces on one quantum, and most of it is the smallness of the electron's energy against its unit of charge (S.44). The units are the electron's: one interval is $\omega_e\hbar/(m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_e$, and one Link is $\sqrt{3}\omega_e\bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_e$. What a click reads of a quantum of frequency ω is its share, $T\sin\omega$ (S.14), at most T , against the electron's rest energy $T\sin\omega_e$. LHAASO's photons of 1.4 PeV [66], read by a calorimeter as a sum of such shares, give $\omega_e < 3.7 \times 10^{-10}$, $\text{den} > 1.5 \times 10^{19}$ and $\gamma_K - 1 = \text{den}/\text{num} - 1 = \omega_e^2/2$ below 6.7×10^{-20} for every orientation of the lattice (the band's top, $\omega = \pi$ along the diagonal, is a GameBoard statement and no click's energy) [1, what a click reads of a quantum]. The Link's bound of Section 10.2, for a beam along an axis, the Link over the unit, gives $\omega_e < 2.2 \times 10^{-14}$ and $\gamma_K - 1 < 2.4 \times 10^{-28}$ (DERIVED, two derivations; CLICKS; S.24, S.27). Γ reads nothing of ω_e or of α , and nothing of the clocks bounds it: the clocks read the mean of the write's dithered sub-unit level, the exact rest, and not Γ (an argument on the mean; the dynamics under a dithered level are open; S.26). No click of the clocks bounds Γ . The gap bound binds the matter family's own denominator, $\text{den} > 1.5 \times 10^{19}$ for the least pair; it binds Γ only in a file that writes that pair over Γ , as the shipped file does. The energy line ties it to T , E_s and k_w (CLICKS) [1, a body and its light keep one energy; the law's alpha is a coefficient of the file; the units are the electron's and the band's top is the highest photon energy; what is open, item 19][3, the strong force's kernel derived: the colour coupling at the Link is the charge's alpha, no running, not the strong force].

10.2 Nature's numbers enter only as clicks

What the lattice fixes with no number. Before any file, the lattice fixes the following. The geometry, with its group of 48, hence one preferred frame and $\alpha_1 = -8$. The dynamics, reversible to the bit with no loops, hence no running of α and no Lamb shift. The conservations per family. The band, with light at $1/\sqrt{3}$ Link per interval under the causal bound 1. The paces, the exponential metric with the finest level $1/\Gamma$. The holders' kernel $3/(4\pi r)$. The click layer's bilinear credit. And one scale Γ with pure numbers of order one, every large number being a power of Γ or a declaration, each with its status in S.54. In one sentence: the lattice is a machine of forms; it fixes every form and every ratio of order one, conserves, reverses and makes no noise; it fixes no strength, no scale and no large number; those are the file's [1, the properties of the lattice].

The calibration series. Every measured number of an experiment is first written as a click pattern on a declared NodeReader against a declared clock. The families' pairs and the units Γ and T are fixed by that match, and by nothing read off the GameBoard. The two ratios the model fixes without it are light's speed of $1/\sqrt{3}$ Links per interval and the causal bound of 1 (ASSUMPTION, CLICKS) [1, the reading; nature's numbers enter the board only as clicks]. No coefficient of the file is computed from the lattice. Each is read from one measurement of nature on a bound body; the lattice fixes the form of the reading, and the file holds the integer, Table 4 (S.54). No number comes from the lattice itself (HYPOTHESIS for any such number, S.54) [1, the calibration series]. One check stands beside the table, the two clocks. A cavity of bodies and

Table 4 The calibration series: each coefficient of the universe’s file, the one measurement of nature that reads it, and the layer it lives in (DERIVED for every form of reading, the integer the file’s). The lattice reads none of them.

The coefficient	The one click that reads it	The layer
T , the unit of action	Planck’s constant: the counter’s unit of credit and Millikan’s slope (Section 7.4)	the click
The Link and the interval	ω_e ’s bounds from the highest photon, LHAASO’s 1.4 PeV, and light’s dispersion, MAGIC’s bound [67] (S.24, S.27)	the board, through the electron’s family
Γ , the board’s clock	bounded by no click of the clocks (Section 10.1); the energy line ties it to T , E_s and k_w (S.26)	the board
The gaps, one per family	the masses: ω_e by the electron’s mass, $g_p/g_e = (m_p/m_e)^2$ under one denominator, read as click masses $T \sin \omega_s$	the board, through bound bodies
E_g , gravity’s weight	Newton’s G , the fall and the orbits, Eq. (20)	the board, through the fall
k , the sign holder’s weight	Coulomb’s force between two quanta, $k/(\Gamma E_s) = 1/28.33$ (S.26)	the board, through two free quanta
c_{vac} , the vacuum content	declared 0; no click reads it, every row reading the content; GW170817 bounds $\gamma_{\text{GW}} - \gamma_{\text{EM}}$	the board
The nuclear holders’ ranges and weights	the nuclear binding and the nuclei’s sizes, if the holders are admitted (Section 10.3; HYPOTHESIS)	the board, through the nucleus

an atom’s beat shift alike by N at long wavelength, as nature’s clocks do, atom against atom and cavity against maser [38], with a cavity of declared faces at N^2 (DERIVED; CLICKS; S.31).

One prediction, untested, one check and one open question. One number follows from the law’s lines where general relativity and quantum theory do not meet. It is entered as an explicitly named hypothesis, and no click has read it yet (DERIVED, HYPOTHESIS; S.41): *the gravity of light* (CLICKS, untested). A light packet pulls at $1 \times E/c^2$, where general relativity’s pencil pulls at $2 \times E/c^2$ [69]. The check: a bound body laid as one record on two paths at two heights has its beat scale as $N = 1 - U$, so the visibility of the fringes falls as $|\cos(\Omega \Delta U T/2)|$. This is 0.989 for a strontium clock at one metre for one second, the magnitude of the semiclassical expectation [68], and it is unmeasured (CLICKS; S.41). The open question: a collapse ends in frozen clocks that read nothing, are read, and are dark by the share’s declaration (Section 7.2). Whether such a region radiates, Hawking’s temperature [73] and its entropy are open; the argument for $T_H = 0$ was withdrawn after the stencil’s reading (S.41, Section 10.3).

10.3 Findings against nature and open questions

GameBoard readings, not findings. The rule’s own universe at [2, 3] is no finding against nature, because nature measures at its own pair (GAMEBOARD; Section 8, (e)). The masses are families. Nature’s mass ratios are not whole numbers ($m_p/m_e = 1836.153$), so no particle of nature can be read as a cluster of the electron’s quanta. The proton is one quantum of its own family, and hydrogen’s clicks read it as one unit of W over 0.84 fm (a comparison for the ratios, ASSUMPTION for the families; CLICKS; S.53). What the click’s structure derives of the masses is in S.53: the rest energy $m^* c_m^2$ exactly, the ratio of two declared rotations, the ceiling at LHAASO’s bound and the kinetic excess of a conversion, and no cause for 1,836 (DERIVED, two derivations). Charge universality and the equivalence between families follow from the writes (DERIVED). If the families share one denominator, the masses are the square roots of the gaps (HYPOTHESIS). [3, the proton’s two readings]

The declarations. The model declares the following and does not derive it. This is the law’s own list, and each item carries what is derived once it is declared (ASSUMPTION for every declaration). (1) A family: the name of its row, its pair, its reads and its dimension, and the integers Γ , T and the width (derived: its band, its tensions and the forms of its clicks). (2) A NodeReader: the region of two Nodes or more and half a wavelength, the window, the seed and generator, the floor and the whole (derived: the share, Born’s form, one quantum one click per record, and the uncertainty relation as these three). (3) A NodeReader with a record of its own: the record laid over its Nodes, two or more, its parts the record’s parts, the modes, held by its declaration and never bound by a well, its write at the Node the draw picks (derived: the exchange, the taking, the null window’s write, Zeno’s blind and the telegraph’s blind). (4) The giving’s rate: the beat’s current over W_c is derived, the floor is declared. (5) The conversion’s table and rate: a hypothesis under its own name, declared to happen and derived in what it does. (6) The lay. (7) The receding face, a declaration of the file. (8) The credit’s one draw through the root, the one declared non-local element. (9) The erasing front: the law’s line by two derivations and the model’s decision, built in the implementation of 2026-10-03. The marks of this paper agree with the list one to one. Nothing in the list is marked derived, and what each item gives is marked derived under it [1, the declarations by name; what produces a click is not known][3, the receding face; a detector as nature’s instrument]. *The four forces, as declared.* On the board there are only records stepped by Rule3 and holders written by the hold. On the board the weak force is nothing but the conversion click, a change of family at a click, and the strong force inside a baryon is nothing but one record of three lines read whole; both appear only in clicks and after them. The nuclear binding is a bound body in the binding holder’s well. Electromagnetism and gravity are the holders’ continuous reads and writes, which a body’s standing reads between clicks. The paper’s account of the forces stands under its own name and claims nothing beyond its marks. The weak force is the conversion click (HYPOTHESIS under its own name): a click that takes a quantum of one family and gives it as quanta of others, the rotations adding. It is the act of one Node, with no row of its own and no range but the Node. The neutron’s end is such a click. Its table and rate are declared and read from nature, as the lifetimes are (Section 5.4; item 5). The strong force inside a baryon is one record, and neither a force nor a body. The baryon is the quantum of a family of dimension 3, and the quarks are its three lines; there is no quark family and none is needed, no count per line and no click of one line. Confinement is the record’s indivisibility under the click (DERIVED from its declared row; S.37). The three lines are three real lines stepped by the one rule. They are rotated among themselves only by the colour holder’s phases on each Link (S.37), as the plane’s two lines are by the sign holder’s, and never by the lattice’s rotations. The testable consequence is that the three fractions of a hard click sum to one and no line is reported alone. The baryon number is derived with them; the flavours, the quark masses and the fractional charges are not derived and not claimed; the mesons are named open. The

strong force between nucleons is a bound body in the gapped binding holder's well. Its row [2400, 2401] is declared, and the range $R = 20$ Links and Yukawa's form are derived from its declared row, the gap. The nucleus is a bound body in that holder, as the atom is in the sign holder's. The proton is three planes and the neutron three real lines, two families, with the conversion the change between them; the full nuclear holders are named open (Section 6.4). Electromagnetism is the sign holder, Poisson's rest and the rotation, with k declared. Gravity is the massless holder, the content and the paces, with E_g declared. Their forms are derived from their declared rows, continuous reads and writes of the Nodes' laws (Section 3). Only the weak force is a click, which is why it alone changes a family [1, the forces as the law holds them; the strong force; the nuclear holder].

Findings against nature. The massless holder's travelling events carry the breathing polarisation and the lattice-aligned $+$ and no $\times$. Nature's gravitational waves carry $+$ and $\times$ at every orientation (CLICKS) [1, what is open, item 23]. Gravity's readers are real records with no read of a current, so a gyroscope precesses by 0 from the motion of a rotating body. Gravity Probe B measured 37.2 ± 7.2 milliarcseconds per year, against general relativity's 39.2 [74] (DERIVED, CLICKS) [3, general relativity in the law; the current's read]. A metric with no g_{0i} in its preferred frame has $4\gamma + 4 + \alpha_1 = 0$, so $\alpha_1 = -8$ at $\gamma = 1$. Nature has $|\alpha_1| < 4 \times 10^{-5}$ from the binary pulsar PSR J1738+0333 [75]. This is a finding (DERIVED, CLICKS; S.35) [1, what is open, item 15]. With $g_{0j} = 0$ in the lattice frame, $g_{00} = -e^{-2U}$ and g_{ij} isotropic, which is the standard post-Newtonian gauge, the two g_{0j} coefficients vanish separately. They give $\alpha_1 = -8$ and $\alpha_2 = -1$ together, and their sum is the gauge-invariant statement above. Nature has $|\alpha_2| < 4 \times 10^{-7}$ from the Sun's spin [60] and $< 1.6 \times 10^{-9}$ from solitary pulsars [76]. Like α_1 , this is conditional on the metric read for a static source (named, not claimed). At the second order in z the ball of dust alone decelerates, $q_0 = \Omega_m/2 > 0$, where the supernovae read $q_0 \approx -0.5$, which is flat Λ CDM's $\Omega_m/2 - \Omega_\Lambda$ [53][54][55]. The paper claims nothing beyond $z \ll 1$ there. The law's documents carry, under its own name, the hypothesis that would close it [1, the vacuum writes]; that hypothesis is not this paper's claim (CLICKS).

The lattice's frame, and its mend as a hypothesis. At long wavelength and small gap every record's band is a Lorentz form, $\omega^2 = \omega_0^2 N^2 + c_m^2 k^2$. So the moving clock's γ follows, and no Michelson-Morley shift appears at nature's pair, to 10^{-19} (at the computed pair c and c_m differ, and a matter-armed interferometer would shift at order v^2). But the content's Poisson rest in the lattice's frame gives a moving mass no g_{0a} and no g_{ab} (DERIVED; CLICKS) [1, the lattice's frame is read by gravity alone, and its carrier is a pair of Links; what is open, item 15]. The mend is an explicitly named hypothesis with no line written (HYPOTHESIS): a carrier on a pair of Links, sourced by the mass current. Against it stands this: among the five forms tried, no real, local, explicit, bijective lattice line on Rule3's clock carries g_{0a} (DERIVED for the five forms, no theorem over every form; S.36). [3, nothing that is not Rule3 remains]

The strong force, by derivation. The strong force is two things (the declarations above). One is confinement, with no coupling. The other is the nuclear binding under the declared holders, read by the nucleon families alone; under it the size of the nucleus, its binding and its light are derived in form, and its shells are not. Confinement is not a force (the declarations above). A hard click reads three fractions that sum to one, the lines' shares, and they are unchanged over the momentum transfer: Bjorken scaling is exact, where nature's measurements show logarithmic violations. This is an explicit finding (DERIVED from the declared row; CLICKS; S.37, S.56). No derived body of the law reaches the nuclear binding (CLICKS). A body's binding per quantum is bounded by $0.76 CW/(E)$ at one Node, and is about $0.36 CW/(Er)$ for a uniform ball of r Links, with E the holder's divisor (DERIVED, a bound; GAMEBOARD; S.38). Under the families' reading the nucleon is one quantum whose binding is the nuclear holder's well, $W \geq 2.35 \Gamma \kappa / \omega_p^3$ (S.45). Beside that hypothesis stands a finding of the implementation: at the design's integers the nucleon's fixed point in its own holder's well stands on the wide branch, the content $1,217 = 0.20 \Gamma$ at the centre, $2 \cos \omega_b = 1.392$ against the vacuum's 1.333, the period 8; the generator's three-plane rounding lays 43 quanta over 33 Nodes from the design's 25; the dedicated test asserts the fixed point's return and the standing over one period against the twin without the holder (43.7 against 20.6 quanta); the compact nucleon of nature's size stays a hypothesis (GAMEBOARD for the body) [1, the nuclear holder]. The nucleus is therefore a hypothesis under its own name: a bound body of nucleon quanta in the well of a nuclear holder, Yukawa's kernel (S.39) with its range and weight declared, read by the nucleon families alone (Section 7), together with a core holder of shorter range and negative weight; four declared integers per nucleus (HYPOTHESIS). Under it the threshold, the binding at a body's size, fusion and fission with the shape of Bethe and Peierls [46], and the size law are computed in form (one derivation; S.45). The findings are these. One well gives $R \propto 1/A$, where nature's nuclei grow as $A^{1/3}$ and saturate; the core holder is the saturating declaration. The shells and the magic numbers (Pauli's exclusion) and the neutron's lifetime are explicit findings (CLICKS). [1, the strong force; the nuclear holder; the nucleus; what is open, item 1][3, is the strong force not events that do not split].

The weak force. The model has no electroweak theory; its weak interaction is the conversion click (the declarations above). Every rotation holder, massive or not, couples at α at the Link; its gap is its range and never its strength (DERIVED; GAMEBOARD; S.39). No G_F follows. A three-line record read as (ν, e) under one $O(3)$ has no fourth rotation, so no Z , where nature has the neutral current [79] and the Z [80]. Parity is conserved, where nature's weak interaction violates it [81]. And the law has no spontaneous click, so the conversion has no rate and the neutron's 880 s has no source. These are three explicit findings (CLICKS). What decays in the model is a body and never a quantum (Section 7), so the neutron's end is the conversion [3, the weak-force reading]. The conversion is built as the fifth list of the one click act: a record is converted whole at its Node at the weights [window, rate - window], and the records out are laid by the count at the massless pair. The neutron's file, read beside its blind with its three named findings, stands in S.62, a reading of the implementation and no claim. The table and the rate are nature's numbers, and nothing of the neutron's mass or its excess is claimed (ASSUMPTION for the table and the rate; CLICKS; S.56 (3)) [1, the click writes on the GameBoard; what the law declares and does not derive][2, the input].

The gaps in two columns. The first column is the file's, and the method invites another file: α through k , G through E_g , the Link and the masses, the hierarchy of the two weights, the spin holder and the click's unit. The second column is the rule's. Each of its gaps has its root in the findings above, and none is the fault of a number [3, why the gaps]: no loops

(α 's running [80], the Lamb shift, $g - 2$), no spontaneous click (the lifetimes), no vector for the exclusion, no line for g_{0a} , and a step that commutes with the cube's reflections (the Z and parity).

The record's parts are its particles. An explicitly named hypothesis beside the families' reading (S.56 (7)) and the account of the forces, not the law; the engine's gate decides it (HYPOTHESIS; CLICKS). Nature's quanta are the parts of the matter records and the records of the three holders. The baryons are the parts of the three-line record (S.37). The charged leptons are families of the plane record, and the neutrino is a real-line family, neutral by the law's own line. They are converted into each other only at the conversion click, by a declared table at a declared rate, read from nature as the lifetimes are. The neutrino's record meets no transition of the bodies that stand for nature's counters except through the conversion, and a declared NodeReader of its family would click it. Postulate 2 holds. Its test is the neutron's end, the conversion of one part at a declared rate, built and run in the neutron's file as the weak force's paragraph states. It gives neither the rates nor the mixing nor parity's violation, and this section's findings stand [1, the hypotheses under their own names].

Open. The hypothesis that only a click moves a quantum between families (Section 5.4; HYPOTHESIS), with its gate files below. No standing body is computed in this version of the implementation; the three problems of its computation and the fix are named (Section 7.3). The black hole's quanta (S.41). The shelved ion's telegraph [70]: bimodal counts with exponential dark periods. The law without the NodeReaders' writes fails it (Poisson at half the rate), and the clicks of a NodeReader with the ion's own record over its Nodes meet it. The shelving is the taking click of the weak drive's quantum, at the Zeno-limited rate Ω_w^2/R [72]. The return is the giving click, at a declared rate. The bright periods have the mean R/Ω_w^2 , and the dark periods the declared τ_D . The shape of the statistics is derived from the known form of the rate equations [71] (DERIVED under a declaration; the rate itself a HYPOTHESIS under its own name; S.56 (15)). The implementation's versions of the click's write, of the giving as a source in time (its energy closed by two derivations), of the resonant act, of the hole of a dense record and its exact factor, of the dark's hazard and of the packet on the open board are recorded in S.62, each with its reading beside its blind: the Zeno column, the resonance file, the anticoincidence and the telegraph's refusal. Each is a computed reading or a named finding, and no claim of nature. They change no number of Sections 9.2 and 9.4. The three gate files of the click's write are the ion's body with its modes as parts and two declared rates, the anticoincidence's (S.57) and the Zeno file's (S.59). They are derived here as the blind rows of their files, and a reading that misses its blind is a named finding. A gate-built register stands outside by postulate 2 (Section 3.6). The model does not derive from the Nodes' laws where a click happens. It declares where, a NodeReader, with or without a record of its own, and derives what the click does once declared. A bound body of nature that jumps with no declaration does nothing but Rule3; this is the method's widest open gap. A spread body's click, the exchange at its entry Node and its settling a reading, stands with it. The click's picture is closed in its form and open in its cause. What is open is this: the draw, which meeting is realised and at which Node; where a click happens, declared; the conversion's table and rate; the resonance against the share at the computed pair; and the engine's build, the carried remainder of the credit's window with its own blind on N and the readings of the three files of a NodeReader with its own record through the one act. The form is closed and claimed; the cause is open and not claimed. A single quantum's track of thousands of ionisations, as a cloud chamber shows it, is not derived and not claimed; a click takes one whole quantum of the record. The click's write conserves no momentum, since the quanta are laid with a phase and no wave number. So the recoil is the NodeReader's, where nature's counter recoils, as in the Mössbauer line and the Compton shift: a named finding beside what produces a click. The empty wave of a clicked record still writes its form into the holders it sources until the front reaches it, and beyond the click's light cone it is never erased. This is a price of locality, beside Section 5.7. The front is built in the implementation; its gate is the three files' numbers bit for bit and the back-in-time check's MATCH across it (S.62). Also open: an exactly conserved integer charge; gravity's divisor E_g (EXPERIMENT); the matter pair's gap; what fixes Γ ; and a current's read for real records [1, what is open]. The relations among the file's integers are the next paper's program, each an explicitly named hypothesis tested by clicks: one denominator for all the families, and whether k , E_g , T and Γ are tied, with the energy line of Section 10.1 already a law (HYPOTHESIS) [1, no free number but the mass]. The atom: the law does not support the reduced mass, since a Node holds no joint amplitude of two records (postulate 2 of Section 3.6) and the nucleon's centre is one classical variable under line (2) of S.60. So the isotope shifts and the muonic atoms are outside what it derives. The many-body correlation is not reproduced by a product of records (helium 1.4 percent, positronium 56 percent, GameBoard readings; S.60), and Pauli's exclusion has no line (S.46, S.60) [1, the atom is a bound body of the holder of the sign].

10.4 What can follow from here, and how to compute it

Every sentence of this subsection is inspiration and no claim: a definition in the law's terms, as the law holds it; what could follow; and how one would compute it on the GameBoard, with its blind. Nothing in it is a result, and the paper's one claim stays the method. *The black hole.* The definition. The law's static metric is the exponential one, $N^2 = e^{-2U}$ and $h^2 = e^{2U}$ with $U = m/r$. It has no horizon and no singularity, and the clock p_0 is never 0 at any content, so a collapse ends in a frozen clock and not at a horizon. In the model, a black hole at nature's scale is a deep clock, defined by its light ring at $r = 2m$ and its shadow's capture radius $2em$, the prediction of Section 8 under the one assumption; this adds nothing to it. What could follow. A Node whose content passes $(\Gamma/2) \ln(2\Gamma)$ has its Link's pace rounded to 0: it reads nothing and is read by its neighbours. It is a one-way valve that holds its level and gravitates, has no share and clicks nowhere. This is the law's one-Link black hole, $\Gamma/2$ quanta at a Node, about 10^{-17} kg at the Link's bound; dark matter as frozen matter is a named hypothesis. Whether a frozen Node radiates is open: no line puts the flux through its surface at 0, and none derives Hawking's temperature. Under the composed paces the compact body passes $\Gamma \div 2$ standing and no Link closes, so no body of the generator reaches the frozen content; that content is reached by declaration and not by a collapse. How to compute it. One declared Node at the frozen content inside an active region, with the loader's refusal lifted for that one file and said so. The GameBoard reading of the outside level m/r and of the shares at the neighbours, with the blind of no click at the frozen Node and the neighbours' clicks unchanged. The form's flux through the frozen Node's six Ports over a long run, against the blind 0 and against Hawking's temperature in the law's units, a reading and not a gate, since nature

has no click of that radiation. And the compact body's fixed point at rising count under the composed paces, the clock's depth read against $\Gamma \div 2$, with the blind standing at every count the width admits (inspiration; GAMEBOARD) [1, the paces compose; what is open, item 29; the rows against nature].

11 Conclusion

The mechanism. Universe24 is built from three things. The GameBoard is a cubic lattice of Nodes, each holding two integer levels per family and a remainder, stepped by one line of integer arithmetic, Rule3, which takes a Node's present, its past and its six neighbours through one integer division. The step is reversible to the bit, conserves a quadratic form exactly in the rationals, is covariant under the cube's 48 symmetries, and draws nothing. The NodeReader is a declared region of two Nodes or more with its seed: it reads the inflow through its Ports over a window, draws once by the shares, and writes one quantum at one Node of its region; its draw is the only randomness in the model and the one non-local element. The clicks are the NodeReader's reports, and Universe24 is those clicks: the model's name for what a detector of an experiment registers, defined by the model and not by nature. A click is the meeting of the future, the record run forward from its root, with the past, the transition that waits for it read backward from the click. The GameBoard computes, the NodeReader is the gate, the clicks are the world of the model, and the formulas the clicks give are set beside nature's.

What was derived from the line alone. With no number of nature: the reversibility, the conserved form with its current, the share identity and the Wronskian as the charge (Section 4); the bands, the lattice wave equation and the Klein-Gordon equation with its Schrödinger limit (Section 3.3); the bound body's stability (Section 7.2); the weak-field forms under one named assumption, the moving clock, de Broglie's and Planck's relations and $E = mc^2$ in the band's form (Section 8); the two slits' row and Bell's, Tsirelson's and GHZ's values under the declared credit (Section 9); and the constants' relations (Section 10).

What is new, and what is recovered. New are the lattice's own forms, which have no continuum counterpart and each of which carries its mark: a family's inertial mass $m^* = 3 \tan \omega_0$ and its light speed $c_m^* = \omega_0 / (3 \tan \omega_0)$, with $E = m^* c_m^2$ exact; the post-Newtonian $\gamma_K = q/p$ and $\beta_K = (q + p)/(2p)$ of a pair $[p, q]$; the exponential metric with $\gamma = 1$ exactly and the shadow's capture radius $2em$, 4.6 percent over Schwarzschild's; $\alpha = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$; the clock's finite-rotation factor $f = 1 + \omega_0^2/12 + \dots$ and the moving clock's fourth-order coefficient 1.6926 at $[2, 3]$; Bell's 478/169 at rational angles; and the click's booking identity with the record's own quantum W_{rec} . Recovered, in lattice form and with the known form beside them, are the wave equation, Lorentz's factor at c_m to the second order, Einstein's weak field in the clock's potential, de Broglie, Planck, Coulomb and Yukawa, Born's rule as a declaration, Tsirelson's bound and Newton's G .

The claim, and its limits. The paper has one claim: the method, a mechanism of GameBoard, NodeReader and clicks that gives formulas which also appear in nature, in the order stated. Everything else in the paper is a supporting finding for that claim, each with its mark and its fence, and none of it is a proof about nature. What is put in is named (the one assumption, the credit chosen with Bell's correlations in view, the file's numbers read from nature); what fails is named with its number, $\alpha_1 = -8$, hydrogen's $1s$ under the self-read, the shelved ion's telegraph (Section 10.3). The form of a click is claimed and its cause is not: why a click happens is the draw, a declaration in the open list, on which no claim rests. Why nature is so, we do not claim to know.

Outlook. Nature's formulas are many, and the paper computes no complete set of them. It samples a few places: the atoms (Sections 7.3 and 10.3; S.46, S.60), gravity and Newton's potential (Section 8), and the two slits and Bell (Section 9). Some derivations meet nature's measured values and some miss, and the misses are items of Section 10.3 with their roots where known. The open list is the program, each item a derivation to complete or a gate file to run. The paper gives the direction and not the whole, so that others can take the method, with this set of rules or another, and continue.

Statements and Declarations

Funding. None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data and code availability.** The law's document, the implementation and the scripts of the figures are developed in the public Git repository <https://github.com/Closer24/Universe24>. They are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [43], under the MIT licence. **Author contributions.** The author laid down every postulate and decision of the model, directed the work and wrote the paper. **The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author's direction, for the software, the computations, the derivations and their checks, and the drafting of the text. The model is not an author. The author reviewed every statement and takes full responsibility for the content.

References

- [1] The law of the GameBoard, docs/ALGEBRA.md of the archive [43], cited by the name of its section: the GameBoard and the families (the objects, a family's declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record's share, the click is the detector's report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, with no waves on the event board, what produces a click is not known and nature's numbers enter the board only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.
- [2] The engine, docs/ENGINE.md of the archive [43]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, docs/HIGHLIGHTS.md of the archive [43]: the model's decisions in force, one line each, cited by the line's name.
- [4] K. Zuse, Rechnender Raum (Vieweg, Braunschweig, 1969); English translation: Calculating Space, MIT Technical Translation AZT-70-164-GEMIT (MIT Project MAC, Cambridge, 1970).
- [5] E. Fredkin and T. Toffoli, Conservative logic, International Journal of Theoretical Physics 21, 219 (1982), doi:10.1007/BF01857727.
- [6] N. Margolus, Physics-like models of computation, Physica D 10, 81 (1984), doi:10.1016/0167-2789(84)90252-5.
- [7] T. Toffoli and N. Margolus, Invertible cellular automata: a review, Physica D 45, 229 (1990), doi:10.1016/0167-2789(90)90185-R.
- [8] S. Wolfram, A New Kind of Science (Wolfram Media, Champaign, 2002).
- [9] G. 't Hooft, Duality between a deterministic cellular automaton and a bosonic quantum field theory in 1+1 dimensions, Foundations of Physics 43, 597 (2013), doi:10.1007/s10701-013-9702-z.
- [10] G. 't Hooft, The Cellular Automaton Interpretation of Quantum Mechanics, Fundamental Theories of Physics 185 (Springer, Cham, 2016), doi:10.1007/978-3-319-41285-6.
- [11] I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, Physical Review D 49, 6920 (1994), doi:10.1103/PhysRevD.49.6920.
- [12] D. A. Meyer, From quantum cellular automata to quantum lattice gases, Journal of Statistical Physics 85, 551 (1996), doi:10.1007/BF02199356.

- [13] P. Arrighi, An overview of quantum cellular automata, *Natural Computing* 18, 885 (2019), doi:10.1007/s11047-019-09762-6.
- [14] T. Farrelly, A review of quantum cellular automata, *Quantum* 4, 368 (2020), doi:10.22331/q-2020-11-30-368.
- [15] G. M. D'Ariano and P. Perinotti, Derivation of the Dirac equation from principles of information processing, *Physical Review A* 90, 062106 (2014), doi:10.1103/PhysRevA.90.062106.
- [16] R. P. Feynman, Simulating physics with computers, *International Journal of Theoretical Physics* 21, 467 (1982), doi:10.1007/BF02650179.
- [17] J. A. Wheeler and R. P. Feynman, Interaction with the absorber as the mechanism of radiation, *Reviews of Modern Physics* 17, 157 (1945), doi:10.1103/RevModPhys.17.157.
- [18] J. G. Cramer, The transactional interpretation of quantum mechanics, *Reviews of Modern Physics* 58, 647 (1986), doi:10.1103/RevModPhys.58.647.
- [19] Y. Aharonov, P. G. Bergmann and J. L. Lebowitz, Time symmetry in the quantum process of measurement, *Physical Review* 134, B1410 (1964), doi:10.1103/PhysRev.134.B1410.
- [20] Y. Aharonov and L. Vaidman, The two-state vector formalism: an updated review, *Lecture Notes in Physics* 734, 399 (Springer, Berlin, 2008), doi:10.1007/978-3-540-73473-4_13.
- [21] H. Price, *Time's Arrow and Archimedes' Point: New Directions for the Physics of Time* (Oxford University Press, New York, 1996).
- [22] K. B. Wharton and N. Argaman, Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics, *Reviews of Modern Physics* 92, 021002 (2020), doi:10.1103/RevModPhys.92.021002.
- [23] N. Harrigan and R. W. Spekkens, Einstein, incompleteness, and the epistemic view of quantum states, *Foundations of Physics* 40, 125 (2010), doi:10.1007/s10701-009-9347-0.
- [24] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physica Physique Fizika* 1, 195 (1964), doi:10.1103/PhysicaPhysiqueFizika.1.195.
- [25] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), doi:10.1103/PhysRevLett.23.880.
- [26] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Physical Review Letters* 48, 291 (1982), doi:10.1103/PhysRevLett.48.291.
- [27] P. M. Pearle, Hidden-variable example based upon data rejection, *Physical Review D* 2, 1418 (1970), doi:10.1103/PhysRevD.2.1418.
- [28] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein-Podolsky-Rosen experiment, *Physical Review A* 47, R747 (1993), doi:10.1103/PhysRevA.47.R747.
- [29] A. Garg and N. D. Mermin, Detector inefficiencies in the Einstein-Podolsky-Rosen experiment, *Physical Review D* 35, 3831 (1987), doi:10.1103/PhysRevD.35.3831.
- [30] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), doi:10.1038/nature15759.
- [31] M. Giustina et al., Significant-loophole-free test of Bell's theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), doi:10.1103/PhysRevLett.115.250401.
- [32] M. A. Rowe, D. Kielpinski, V. Meyer, C. A. Sackett, W. M. Itano, C. Monroe and D. J. Wineland, Experimental violation of a Bell's inequality with efficient detection, *Nature* 409, 791 (2001), doi:10.1038/35057215.
- [33] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), doi:10.1103/PhysRevLett.115.250402.
- [34] S. Storz et al., Loophole-free Bell inequality violation with superconducting circuits, *Nature* 617, 265 (2023), doi:10.1038/s41586-023-05885-0.
- [35] T. Regge, General relativity without coordinates, *Il Nuovo Cimento* 19, 558 (1961), doi:10.1007/BF02733251.
- [36] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* 59, 521 (1987), doi:10.1103/PhysRevLett.59.521.
- [37] P. Touboul et al. (MICROSCOPE Collaboration), MICROSCOPE mission: final results of the test of the equivalence principle, *Physical Review Letters* 129, 121102 (2022), doi:10.1103/PhysRevLett.129.121102.
- [38] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), doi:10.12942/lrr-2014-4.
- [39] J. Collins, A. Perez, D. Sudarsky, L. Urrutia and H. Vucetich, Lorentz invariance and quantum gravity: an additional fine-tuning problem?, *Physical Review Letters* 93, 191301 (2004), doi:10.1103/PhysRevLett.93.191301.
- [40] D. Mattingly, Modern tests of Lorentz invariance, *Living Reviews in Relativity* 8, 5 (2005), doi:10.12942/lrr-2005-5.
- [41] R. Courant, K. Friedrichs and H. Lewy, Über die partiellen Differenzgleichungen der mathematischen Physik, *Mathematische Annalen* 100, 32 (1928), doi:10.1007/BF01448839.
- [42] H. Yilmaz, New approach to general relativity, *Physical Review* 111, 1417 (1958), doi:10.1103/PhysRev.111.1417.
- [43] A. Gonen, Universe24: the implementation of one integer rule on a cubic lattice, read through clicks, version 1.0.0, Zenodo (2026), doi:10.5281/zenodo.22738746 (the concept DOI, resolving to the latest version).
- [44] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), doi:10.1007/BF01397477.
- [45] P. Grangier, G. Roger and A. Aspect, Experimental evidence for a photon anticorrelation effect on a beam splitter: a new light on single-photon interferences, *Europhysics Letters* 1, 173 (1986), doi:10.1209/0295-5075/1/4/004.
- [46] H. A. Bethe and R. Peierls, Quantum theory of the dipion, *Proceedings of the Royal Society A* 148, 146 (1935), doi:10.1098/rspa.1935.0010.
- [47] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), doi:10.11429/ppmsj1919.17.0.48.
- [48] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), doi:10.3847/2041-8213/aa920c.
- [49] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), doi:10.1063/1.1704233.
- [50] S. I. Pohozaev, Eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$, *Soviet Mathematics Doklady* 6, 1408 (1965).
- [51] S. I. Pekar, *Untersuchungen über die Elektronentheorie der Kristalle* (Akademie-Verlag, Berlin, 1954).
- [52] W. H. McCrea and E. A. Milne, Newtonian universes and the curvature of space, *The Quarterly Journal of Mathematics* os-5, 73 (1934), doi:10.1093/qmath/os-5.1.73.
- [53] A. G. Riess et al., Observational evidence from supernovae for an accelerating universe and a cosmological constant, *The Astronomical Journal* 116, 1009 (1998), doi:10.1086/300499.
- [54] S. Perlmutter et al., Measurements of Omega and Lambda from 42 high-redshift supernovae, *The Astrophysical Journal* 517, 565 (1999), doi:10.1086/307221.
- [55] D. Brout et al., The Pantheon+ analysis: cosmological constraints, *The Astrophysical Journal* 938, 110 (2022), doi:10.3847/1538-4357/ac8e04.
- [56] R. V. Pound and G. A. Rebka, Apparent weight of photons, *Physical Review Letters* 4, 337 (1960), doi:10.1103/PhysRevLett.4.337.
- [57] B. Bertotti, L. Iess and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* 425, 374 (2003), doi:10.1038/nature01997.
- [58] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), doi:10.1103/PhysRevLett.13.789.
- [59] S. Coleman and S. L. Glashow, High-energy tests of Lorentz invariance, *Physical Review D* 59, 116008 (1999), doi:10.1103/PhysRevD.59.116008.
- [60] K. Nordtvedt, Probing gravity to the second post-Newtonian order and to one part in 10^7 using the spin axis of the Sun, *The Astrophysical Journal* 320, 871 (1987), doi:10.1086/165603.
- [61] T. Jacobson, S. Liberati and D. Mattingly, A strong astrophysical constraint on the violation of special relativity by quantum gravity, *Nature* 424, 1019 (2003), doi:10.1038/nature01882.
- [62] F. W. Stecker, Limiting superluminal electron and neutrino velocities using the 2010 Crab Nebula flare and the IceCube PeV neutrino events, *Astroparticle Physics* 56, 16 (2014), doi:10.1016/j.astropartphys.2014.02.007.
- [63] The Event Horizon Telescope Collaboration, First Sagittarius A* Event Horizon Telescope results. VI. Testing the black hole metric, *The Astrophysical Journal Letters* 930, L17 (2022), doi:10.3847/2041-8213/ac6756.
- [64] The Event Horizon Telescope Collaboration, First M87 Event Horizon Telescope results. VI. The shadow and mass of the central black hole, *The Astrophysical Journal Letters* 875, L6 (2019), doi:10.3847/2041-8213/ab1141.
- [65] S. Hossenfelder and T. Palmer, Rethinking superdeterminism, *Frontiers in Physics* 8, 139 (2020), doi:10.3389/fphy.2020.00139.
- [66] Z. Cao et al. (LHAASO Collaboration), Ultrahigh-energy photons up to 1.4 petaelectronvolts from 12 gamma-ray Galactic sources, *Nature* 594, 33 (2021), doi:10.1038/s41586-021-03498-z.
- [67] V. A. Acciari et al. (MAGIC Collaboration), Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C, *Physical Review Letters* 125, 021301 (2020), doi:10.1103/PhysRevLett.125.021301.
- [68] M. Zych, F. Costa, I. Pikovski and C. Brukner, Quantum interferometric visibility as a witness of general relativistic proper time, *Nature Communications* 2, 505 (2011), doi:10.1038/ncomms1498.
- [69] R. C. Tolman, P. Ehrenfest and B. Podolsky, On the gravitational field produced by light, *Physical Review* 37, 602 (1931), doi:10.1103/PhysRev.37.602.
- [70] J. C. Bergquist, R. G. Hulet, W. M. Itano and D. J. Wineland, Observation of quantum jumps in a single atom, *Physical Review Letters* 57, 1699 (1986), doi:10.1103/PhysRevLett.57.1699.
- [71] R. J. Cook and H. J. Kimble, Possibility of direct observation of quantum jumps, *Physical Review Letters* 54, 1023 (1985), doi:10.1103/PhysRevLett.54.1023.
- [72] W. M. Itano, D. J. Heinzen, J. J. Bollinger and D. J. Wineland, Quantum Zeno effect, *Physical Review A* 41, 2295 (1990), doi:10.1103/PhysRevA.41.2295.
- [73] S. W. Hawking, Particle creation by black holes, *Communications in Mathematical Physics* 43, 199 (1975), doi:10.1007/BF02345020.
- [74] C. W. F. Everitt et al., Gravity Probe B: final results of a space experiment to test general relativity, *Physical Review Letters* 106, 221101 (2011), doi:10.1103/PhysRevLett.106.221101.
- [75] L. Shao and N. Wex, New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars, *Classical and Quantum Gravity* 29, 215018 (2012), doi:10.1088/0264-9381/29/21/215018.
- [76] L. Shao, R. N. Caballero, M. Kramer, N. Wex, D. J. Champion and A. Jessner, A new limit on local Lorentz invariance violation of gravity from solitary pulsars, *Classical and Quantum Gravity* 30, 165020 (2013), doi:10.1088/0264-9381/30/16/165020.
- [77] D. M. Greenberger, M. A. Horne and A. Zeilinger, Going beyond Bell's theorem, in *Bell's Theorem, Quantum Theory and Conceptions of the Universe*, ed. M. Kafatos (Kluwer, Dordrecht, 1989), 69, doi:10.1007/978-94-017-0849-4_10; N. D. Mermin, Extreme quantum entanglement in a superposition of macroscopically distinct states, *Physical Review Letters* 65, 1838 (1990), doi:10.1103/PhysRevLett.65.1838.
- [78] B. S. Cirel'son, Quantum generalizations of Bell's inequality, *Letters in Mathematical Physics* 4, 93 (1980), doi:10.1007/BF00417500.
- [79] F. J. Hasert et al., Observation of neutrino-like interactions without muon or electron in the Gargamelle neutrino experiment, *Physics Letters B* 46, 138 (1973), doi:10.1016/0370-2693(73)90499-1.
- [80] The ALEPH, DELPHI, L3, OPAL and SLD Collaborations, Precision electroweak measurements on the Z resonance, *Physics Reports* 427, 257 (2006), doi:10.1016/j.physrep.2005.12.006.
- [81] C. S. Wu, E. Ambler, R. W. Hayward, D. D. Hoppes and R. P. Hudson, Experimental test of parity conservation in beta decay, *Physical Review* 105, 1413 (1957), doi:10.1103/PhysRev.105.1413.